Ontological Rigidity and Universal Single-Mother Formation

Axiom-Free Exclusion of Unaccounted Differences, Causal Admission, and Unified Evolution

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Formal verification and edition note. Formal proofs in this work are verified in the Lean 4 theorem prover [9] (equipped with a small trusted kernel and exportable reference checking, §5); this paper's own audit receipts and automated checks establish a strictly two-tiered axiom closure: the universal authority layer governing single-constructor occurrence unfolding, the root-read identity, lawful world admission, positive total reality, observer-quotient safety, and categorical initial-coface revision is a **zero-axiom constructive kernel** (T0, `#print axioms = []`, invoking neither excluded middle, classical choice, nor quotient soundness), whereas the analytic realization layer supporting pointwise uniform completion of the fixed physical mother source $S_\star$, the continuous–quantum unified field instantiation, and multi-stage analytic window continuation has a full dependency closure strictly restricted to the three standard Lean axioms (T1: propositional extensionality `propext`, classical choice `Classical.choice`, and quotient soundness `Quot.sound`), without back-contaminating the T0 kernel and with zero unfinished placeholders (`sorry`) or unsafe declarations anywhere. Positive verification chains and compile-time overreach guards (`fail_if_success`, `#guard_msgs`) pass under pre-registered contracts. This second edition (v2) overhauls the first-edition preprint (*1 Bit Is All You Need*, DOI: 10.5281/zenodo.23437456; the former title paid tribute to Vaswani et al. [5], without inferring any claim regarding neural architectures or scaling laws), integrating the root-read mother law, living-process total reality across revision seams, the single-root compile-time authority firewall, and paid-AST homomorphic and homological continuation (OB12–OB13).

Abstract

Can a real difference that alters world identity or observable outcomes acquire actual standing without any registered structural provenance? Working across a zero-axiom constructive kernel (T0, `#print axioms = []`) and a standard three-axiom analytic completion layer (T1), we prove the answer is no and cash out this binary admissibility criterion in four interlocking foundational theorems. First, inside a T0 kernel whose weak five-predicate difference vocabulary contains no preset "no-magic" axiom field, a single-constructor occurrence tree establishes the root-read identity $\mathsf{RootRead}(o) = o$; distinct lawful world states of any source-native authoritative root, as well as distinct states of any complete living process crossing local terminals and theory revisions, acquire positive structural identity from injective registered occurrences or complete living currents, constructively excluding external supplements and magic bits without circularity while a compile-time single-root firewall blocks eight classes of bypass oracles. Second, at layer T1, pointwise uniform completion of the finite-programme evaluation image of one primitive-event-generated spin-pair mother source $S_\star$ universally forms—with zero added representability premises—every lawful world state across all relation networks, vocabularies, and authority roots, every complete inquiry, and every macro-process on the `Type 0` interface, recovering occurrences, whole-ledger writes, compilations, and arbitrary-depth histories by left-inverse restriction (eliminating any independent second base source on that interface) and directly instantiating the $\mathrm{Spin}^+(1,3) \times \mathrm{SU}(7)$ continuous–quantum unified field.

Third, at every natural-number cofinal admission stop, the literal syntactic residual equals the paid boundary across all finite allowed action words and sites, generates the next endogenous request, and executes strict Noetherian syntax debit without budget refill along the unique debt lineage before delivering the effect value, receipted answer, whole-ledger destinations, and canonical successor; moreover, double-negation extraction recovers the complete paid abstract syntax tree (paid AST) to execute any pre-fixed additive homomorphism across subsequent cofinal stops with full query traces recorded in born stock, while primitive-by-primitive word replay turns each evaluation step into explicit integral stock relations driving chain morphisms, the cochain differential $d^{0,1}$, and coimage differentials in lockstep (and carrying four prior execution traces across conditional long analytic windows). Fourth, whenever an existing law suffers rooted double emptiness of expression and realization fibres, the obstruction itself generates in zero axioms a categorical initial coface $L_+ \sqcup \{\star_\eta\}$ with unique faithful injective factorization, routing migrated prior debts and the new zero-budget direction through one shared nonterminal first write. Every claim ultimately terminates in three layers of mechanically checkable equalities governing structural recovery, paid continuation, and domain interpretation.

Keywords: ontological rigidity; no unaccounted difference; zero-axiom kernel; root-read identity; universal single-mother formation; same-debt syntax payment; paid-AST homological continuation; categorical initial coface

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1 Introduction: Ontological Rigidity Without Unaccounted Differences and the Single-Mother Programme

Every foundational scientific or formal theory eventually confronts a single precipice: **if even one real difference is permitted to enter the world without accounting for any internal generative provenance, on what principled ground can the theory refuse the next arbitrary patch?** In fundamental physics, such unaccounted patches appear as free coupling dials tuned retrospectively to empirical numbers, externally adjoined measurement-collapse switches, or separate underlying sources opened ad hoc to host new phenomena [7]; in computation and accounting systems, they appear as oracle assignments overwriting state without an execution trace, or vocabulary renamings that silently write off unsettled prior obligations [6][8]. Conversely, if one merely places a prohibitory axiom—"no unaccounted differences (no magic)"—at the top of a formal system, the slogan risks collapsing into circularity or ex-post apologetics: whatever can be derived is called internal, and whatever resists derivation is retroactively declared a primitive.

Over the past four decades, four major formal programmes have addressed distinct facets of this problem: (i) in dependent type theory and resource logics, Martin-Löf's intuitionistic type theory [10] provides witness-bearing Σ-projections (while adopting equality reflection on p. 32 of the 1984 notes), Homotopy Type Theory's Univalence Axiom [11] (Axiom 2.10.3) characterizes type equivalence and type equality $(A =_{\mathcal{U}} B) \simeq (A \simeq B)$ within a universe, linear logic [12] governs duplication and discarding of marked formulas via exponentials, and amortised separation logic [13] verifies heap consumption via LP-inferable credit assertions and a fallible environmental `acquire` instruction (§3.6); (ii) in foundations of physics and specification refinement, informational derivations of quantum theory [14] adopt an operational-probabilistic framework with five informational axioms plus purification, constructor theory [15] expresses physical laws through possible/impossible tasks and the interoperability principle (Principle III), and state refinement completeness [16] augments lower-level specifications under regularity conditions with auxiliary history and prophecy variables (whose projection onto original variables preserves visible behaviours) to construct refinement mappings; (iii) in data provenance and incremental computation, provenance semirings [17] factorize positive relational evaluation via polynomial semiring homomorphisms, while static λ-calculus differentiation [18] (Theorem 3.11) and DBSP stream calculus [19] (§4.1 $\mathbb{Z}$-sets, Theorem 3.4 bilinear formula) establish exact finite-difference updates without turning unpaid residuals into self-paying endogenous requests or step-trace cochain differentials; and (iv) in formal theory change, AGM theory [20] operates inside a fixed language—using partial-meet contraction $K \dot{-} p = \bigcap \gamma(K \perp p)$ to relinquish implication of p, and in conflicting revisions applying the Levi identity to contract $\neg p$ before adjoining p—leaving open how to expand a doubly bankrupt language in zero axioms without contracting unsettled prior obligations.

This paper connects all four seams constructively. Our central advance is twofold: **rather than assuming "no magic" as an axiom, modal primitive, or structure field, we derive the emptiness of external supplements and magic bits without circularity inside a zero-axiom constructive kernel (T0) solely from causal occurrence admission and positive difference classification; we then prove at the standard analytic layer**

(T1) that a single primitive-event-generated mother source $S_\star$ universally forms—without adding representability hypotheses or specification auxiliary variables—all lawful worlds, complete inquiries, and evolving macro-processes on the original admission interface, while reducing endogenous requests, same-debt payments, paid-AST homological continuation, and zero-axiom bankruptcy revision to mechanically checkable equalities.

1.1 The Binary Standing Question, Non-Circular Exclusion, and the Four-Pillar Master Theorem

The phrase "one bit" from our first-edition title carries one exact technical meaning throughout this programme (**OB1**): it names the binary standing question—**can an actual difference that alters world identity or downstream observables acquire lawful standing without any registered structural provenance?** Across all source-native structures studied here, we prove the answer is "no" (the propositional structure MagicBitAt is uninhabited). This binary stamp neither compresses cosmic data into a single binary digit (a deterministic constant has strictly zero Shannon entropy [3]) nor posits a terminating Boolean decider over arbitrary external raw types; its entire mathematical force resides in four load-bearing constructive theorems.

Main Theorem 1.1 (Four-pillar master theorem of ontological rigidity and single-mother formation).

1. **Zero-axiom positive total reality, root-read identity, and occurrence standing (T0, `#print axioms = []`; §§2–3, OB1–OB5).** A single-constructor occurrence tree establishes the mother identity $\mathsf{RootRead}(o) = o$: the predecessor patch trace, current patch, and whole-ledger write-back of a registered occurrence are the intrinsic shape of that occurrence itself. For every relation network N, vocabulary V, and source-native authoritative root R, any two unequal lawful world states of R (spanning finite visits, cofinal landings, and post-cofinal successors) possess distinct registered occurrences $\mathsf{occ}(w) \neq \mathsf{occ}(w')$; for every complete living source process $\mathcal{P}$, any two unequal states (including states separated by local faithful terminals or theory-revision seams) possess distinct complete living currents $\iota(s) \neq \iota(s')$. Without any `NoMagic` input field, both structures positively realize $\mathsf{Total}(\mathcal{R})$ and constructively exclude external supplements and magic bits (Theorems 2.5, 2.6, 2.8), while strictly separating occurred rigidity (Theorem 3.1), observer-quotient blindness (Theorem 3.3), and unstored on-demand futures.
2. **Universal three-domain formation from one fixed spin-pair mother $S_\star$ without a second base source (T1; §4, OB6–OB7).** Fix the primitive-event-generated physical spin-pair mother source $S_\star$. On the lowest type universe (`Type 0`) admission interface, with zero caller-supplied representability or address-coverage hypotheses, pointwise uniform completion of the finite-programme evaluation image of $S_\star$ simultaneously forms: (i) **every** lawful world state $w \in \mathsf{Law}(R)$ across **all** N, V, R; (ii) **every** complete inquiry q across **all** N, V; and (iii) **every** native inquiry-engine macro-process P; and recovers via left-inverse restriction from the same witness its occurrences, whole-ledger write-backs, query compilations, activation laws, and runtime histories at every finite depth $n \in \mathbb{N}$ (Theorem 4.2). Because the authority root R

is universally quantified, no independent second base source is needed on the `Type 0` interface. The $\mathrm{Spin}^{+}(1,3) \times \mathrm{SU}(7)$ continuous–quantum unified field invokes this theorem directly on a single `SpinPair` occurrence to realize the nine-field classical solution, the exact finite-$\hbar$ $4\sqrt{2}$ pairing, and macro-step succession (Corollary 4.3, Theorem 4.4).

3. **Endogenous residual requests, well-founded same-debt payment, and paid-AST homological continuation (§§5, 7, OB8, OB10, OB12–OB13).** For every source-generation family $\mathcal{F}$, initial frame I, and programme `cfg`, at each natural-number cofinal admission stop generated by the run, the literal syntactic residual $r_m = a + (-\mathrm{id})(p)$ lies in the same all-future equivalence fibre as the paid boundary $[a] - [p]$ across all finite allowed action words and all sites, generating the next endogenous request; its syntax budget $Q = c(r_m) \geq 2$ enforces a strict debit without refill along the unique debt lineage, delivering upon $(Q-1)$-step suffix settlement the effect value $\epsilon = D_a(\Gamma, \delta)$, receipted answer, old and born row destinations, and canonical successor (Theorems 5.1–5.3). Furthermore, double-negation extraction recovers the complete paid AST from residual syntax to execute any pre-fixed additive homomorphism f across subsequent cofinal stops while recording complete query traces in born stock and installing unified algebraic, combinatorial, topological, and logical faces together with relation and cochain faces (Theorem 5.4); primitive-by-primitive word replay converts every evaluation step into explicit integral stock relations updating chain morphisms τ_1, the cochain differential $d^{0,1}$, coimage differentials, and same-stop physical readbacks in lockstep (Theorem 5.5), and migrates four prior execution traces across conditional long analytic windows (§7, Theorem E.1).
4. **Obstruction-generated zero-axiom initial cofaces and shared first-write debt revision (T0, `#print axioms = []`; §6, OB9).** Whenever an existing theory Θ suffers rooted double emptiness of expression and lawful-realization fibres at an emitted demand row, the failure η itself generates in zero axioms the coordinate extension $C_\eta = L_+ \sqcup \{\star_\eta\}$, preserving all old theorems and satisfying strict categorical initiality with unique faithful injective factorization into every admissible revision (Theorem 6.2); every structurally anchored typed revision Ξ emitted by a fixed inquiry likewise routes, inside the zero-axiom kernel, both migrated prior debt rows and the new zero-budget direction row through one shared nonterminal first write, where the revised root's own classifier issues separate dispositions sharing a single canonical next current (Theorems 6.4–6.5).

Table 1 summarizes the axiomatic tier, constructive capability, and paired negative control of each pillar; Figure 1 depicts how all four branches grow from the same single-constructor root kernel and mother source $S_\star$.

Table 1 Two-tier axiomatic foundation and paired positive/negative controls across the four pillars.

Load-bearing pillar	Axiom tier	Core constructive theorem and capability	Paired counterexample / compile-time negative control	Sections & proofs
I. Total reality & root identity	T0 `[]`	$\mathsf{RootRead}(o) = o$; root and cross-revision process total reality derive `noMagicBit`; observer safety	Raw Boolean ghost pair; 8-surface `#guard_msgs` firewall; global integral torsion class invisible to all finite complex readings	§§2–3; App. A
II. Single-mother formation	T1 Std3	Completed $S_\star$ programmes universally form worlds, inquiries, and processes across $\forall N, V, R$ with left-inverse recovery	Non-bypass audit on 5 downstream checks; omitting Dirac swap S shifts dark-moment current from $+1/2$ to $-1/2$	§4; App. B
III. Paid & cochain continuation	T0/T1	All-future residual generates request and strict debit; double negation recovers paid AST for method f; step replay updates $d^{0,1}$	Zero numerical effect $\epsilon = 0$ still pays 2 syntax steps; endpoint value equality never substitutes for step-trace identity	§§5, 7; Apps. C, E
IV. Initial-coface revision	T0 `[]`	Rooted double bankruptcy generates initial coface $L_+ \sqcup \{\star_\eta\}$ and unique injection; shared first write	Recovery API seals off post-hoc failure-row submission; refuted equations lack obstruction events and cannot enter revision	§§6, 8; App. C.5

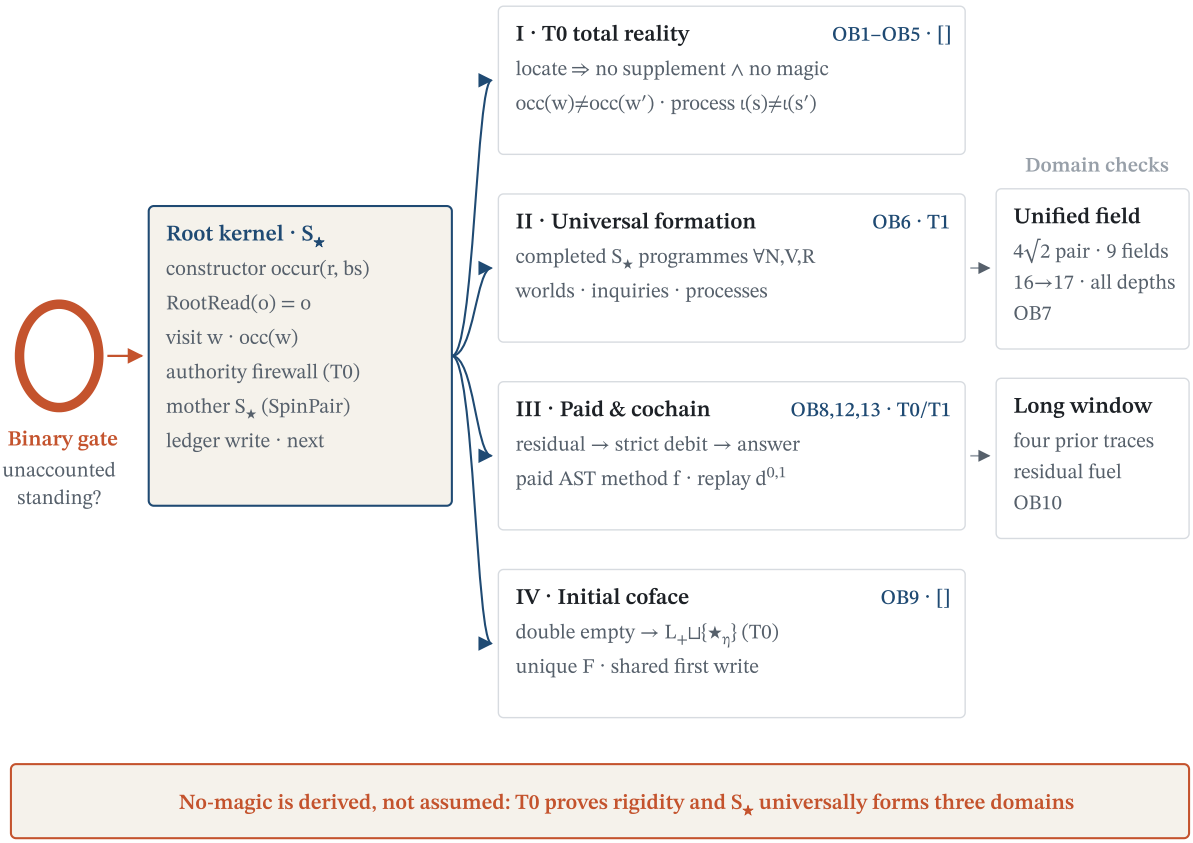


Figure 1 | Single-root occurrence and the four-pillar universal formation forest. The vermilion ellipse on the left marks only the binary entry question (whether an unaccounted difference can acquire lawful standing). Load-bearing construction begins in the deep-blue root kernel ($\mathsf{RootRead}(o) = o$, temporal visit occurrence $\mathsf{occ}(w)$, and the T0 single-root firewall) together with the primitive-event-generated spin-pair mother $S_\star$. Four branches are each paid by their own positive construction: zero-axiom classification derives root and living-process total reality (OB1–OB5); pointwise completion of $S_\star$ programmes universally forms worlds, inquiries, and macro-processes (OB6); all-future residuals drive strict same-debt debit, paid-AST homomorphic action f, and primitive-replay cochain differentials $d^{0,1}$ (OB8, OB12–OB13); and rooted double bankruptcy generates a zero-axiom initial coface routing old debts and the new direction through one shared first write (OB9). On the right, the unified field (OB7) and long analytic window (OB10) invoke their respective branches without bearing their foundational load.

1.2 Three-Minute Pen-and-Paper Anchor: Ghost Projection, Nonzero Square Bill, and Two-Step Revision

Before introducing general dependent types, the reader can verify three miniature calculations by hand in three minutes. Free of black-box notation, they exhibit the exact mechanisms of §§2, 5, and 6.

Example 1.2 (Three-minute pen-and-paper check).

(a) Why one extra raw Boolean bit cannot create a second lawful world (§2.2). Let the initial visit of an authoritative root $R_\circ$ define the lawful world state $w_1 = \mathrm{finite}(v_1)$, and let one compiler successor step define $w_2 = \mathrm{finite}(v_2)$. On the raw product carrier $G = \mathsf{Law}(R_\circ) \times \mathrm{Bool}$, set $g_0 = (w_1, \mathrm{false})$ and $g_1 = (w_1, \mathrm{true})$. The second projection immediately gives $g_0 \neq g_1$; yet lawful

admission accepts solely the first projection $\mathrm{pr}_1 : G \to \mathsf{Law}(R_\circ)$, so

$$\mathrm{pr}_1(g_0) = w_1 = \mathrm{pr}_1(g_1), \qquad \mathsf{occ}(\mathrm{pr}_1(g_0)) = \langle w_1, \varepsilon_{R_\circ}(w_1)\rangle = \mathsf{occ}(\mathrm{pr}_1(g_1)).$$

Conversely, w_1 and w_2 carry distinct reachability constructors (initial vs. one-step), so $w_1 \neq w_2$, and projecting the first component of $\mathsf{occ}(w) = \langle w, \varepsilon(w)\rangle$ yields $\mathsf{occ}(w_1) \neq \mathsf{occ}(w_2)$. Both the raw pair inequality and the single admitted world are genuine mathematical facts; what is blocked is the unlicensed promotion of an unregistered Boolean into a second world.

(b) How a nonzero square bill settles prior debt and feeds its paid AST into the next method (§§5.1–5.4). Over the integers or reals, take the nonlinear expression $a = x \cdot x$, prior environment $\Gamma(x) = 2$, and increment $\delta(x) = 2$ (new environment $x = 4$), with prior execution settled at $p = \mathrm{const}\, 4$. The next endogenous request is the literal residual $r = a + (-\mathrm{id})(p)$:

$$\mathrm{ev}_{\Gamma+\delta}(r) = 4^2 - 4 = 12, \qquad D_a(\Gamma, \delta) = 2 \cdot 2 \cdot 2 + 2^2 = 12.$$

Counting syntax nodes (variable 1, constant 0, unary negation +1, bilinear +1) gives $c(a) = 3$, $c(p) = 0$, and total request budget $Q = c(r) = 3 + 0 + 2 = 5$. After the birth step consumes 1 node, the remaining budget $Q - 1 = 4 > 0$ forces a genuine paid step $(4 \to 3)$ before $(Q-1)$-step suffix reduction reaches budget 0 and delivers $\epsilon = 12$. Crucially, inside the resulting two-layer residual syntax, the extractor `retainedCode` applies double negation $-(-\mathrm{const}\, 12)$ to recover the paid value 12 at zero environment increment, feeds it at the next cofinal stop into any pre-fixed homomorphism such as $f(u) = 3u$ to output $(36, 0)$, and records the three-step primitive reduction chain $[x \cdot x] \to [4 \cdot x] \to [4 \cdot 4] \to [16]$ whose telescoping stock boundary is $[x \cdot x] - [16]$.

(c) How a two-step bankruptcy revision expands the language without erasing prior debt (§6.2). Suppose an old world has support $\{\mathrm{false}, \mathrm{true}\}$, each carrying an open obligation of budget 2, while the old theory has doubly empty expression and realization fibres at true. Step 1 triggers revision: the zero-axiom minimal coface adjoins a preimage-free point $\star_\eta$ at budget 0, producing the initial ledger "migrated old row (budget 2), new row (budget 0)". Taking natural numbers as currents with native write $n \mapsto n + 1$, the first write $0 \to 1$ carries both rows ($2 \to 2$ and $0 \to 0$) to first-write target visit 1, where the revised root's compiler disposes of each row and advances via $n \mapsto n + 1$ to the shared next current 2; step 2 answers at current 2 and advances to 3. Language capacity expands while the prior obligation of budget 2 crosses the seam intact.

1.3 Fifteen-Minute Verification Fast-Tracks by Discipline and Core Notation

Because this paper bridges mathematical logic and dependent type theory, foundational physics unification, programming-language semantics and homological algebra, and philosophy of science, specialists can enter directly along four self-contained routes after Example 1.2:

- **Mathematical logic, dependent type theory, and formal verification**: read **§§2, 3.1, 6.1, and Appendix A** to inspect how the T0 zero-axiom kernel (`#print axioms = []`) establishes $\mathsf{RootRead}(o) = o$ on a single-constructor inductive tree, derives `noMagicBit` constructively without excluded middle or `NoMagic` fields, seals eight overreach surfaces via

compile-time `#guard_msgs` guards, and proves zero-axiom categorical initiality and unique factorization of minimal cofaces.

- **Theoretical physics and continuous–quantum unification**: read **§§4, 3.2, and Appendix B** to verify how pointwise uniform completion of the primitive-event-generated `SpinPair` mother $S_\star$ universally forms all worlds, inquiries, and macro-processes across $\forall N, V, R$ on `Type 0` (excluding any independent second base source), and how the unified field realization recovers the nine-field classical solution, the $4\sqrt{2}$ pairing, and the dark-moment Dirac-swap negative control on the same occurrence.
- **Programming languages, incremental computation, and homological algebra**: read **§§5, 7, and Appendices C, E** to check invariance of the all-environment zero submodule Z under pair lifting, strict Noetherian syntax debit without refill, double-negation paid-AST recovery driving continuing homomorphic actions (`OB12`), and `WordExecution` primitive step replay driving stock boundaries, τ_1, and the cochain differential $d^{0,1}$ (`OB13`).
- **Philosophy of science, metaphysics, and methodology**: read **§§1.1, 2.3–2.5, 3, and 8** to examine how registered occurrence identity realizes sufficient reason constructively without theological regress, how separating occurred rigidity, observer blindness, and unstored future responsibility dissolves deterministic fallacies, and how three-layer failure attribution anchors falsifiability to explicit equalities.

From §2.4 and §4 onward, three standard dependent-type operations recur: **transport** moves an element across a type equality $A = B$; **heterogeneous equality** ($x \doteq y$) asserts that $x \in A$ and $y \in B$ have equal types $A = B$ and agree after transport; and a **dependent direct sum** (Σ-type or finitely supported direct sum) assembles site-indexed carriers M_j with canonical site embedding ι_j, dependent pair $\langle u, e \rangle$, and first projection pr_1 (or `.1`).

2 The Zero-Axiom Total-Reality Kernel, Root-Read Identity, and Single-Root Firewall

Every definition, lemma, and theorem in this section resides in the **zero-axiom constructive kernel (T0, `#print axioms = []`)**. The order of development is strict: admission comes first, exclusion second. We first exhibit the single-constructor occurrence tree and the root-read identity $\mathsf{RootRead}(o) = o$, defining lawful world states as temporal visits so that recovering a world state from its registered occurrence is a single projection; we next establish the Boolean ghost counterexample and the compile-time single-root firewall; and finally, inside a weak difference vocabulary that never mentions magic, we constructively prove root-level and cross-revision process-level total reality.

2.1 Single-Constructor Unfolding, Root-Read Identity, and Lawful World Admission

Why do formal verification frameworks so often rely on external "audit certificates" bolted onto state transitions after the fact? Because they split state evolution and provenance accounting into two separate data structures. At the base of our T0 kernel (`LivingLawRootedAccountedUnfoldingK`

ernel), an actual occurrence is reduced to a **single-constructor** rooted finite exposure tree:

$$\mathsf{RootedAccountedUnfolding}(\mathcal{X}) \ ::= \ \mathsf{occur}(r : \mathcal{X},\ bs : \mathsf{AccountedBranches}(\mathcal{X})). \tag{1}$$

Parallel branching (`parallelAt`), dependent branching (`jointAt`), the unresolved leaf frontier (`frontier`, satisfying the zero-axiom non-emptiness law $\mathsf{frontier}(o) \neq [\,]$ so unresolved obligations cannot silently vanish), the complete provenance trace (`trace`), associative step unfolding (`advance_assoc`), on-demand finite observation (`observe`, storing no completed table of future exposures), and the universal evaluation fold (`fold`) are all zero-axiom derived operations on this sole constructor; moreover, every candidate evaluator commuting with occur factors uniquely through `fold` (`fold_unique`). Neither a second occurrence constructor nor an external receipt schema is primitive.

Built upon this unfolding kernel over a relation network N and vocabulary V, a **source-native authoritative root** R comprises: (i) an authority source (restructuring ledger source, complete event-inventory admission, law surface, and projection law), where **complete event-inventory admission** requires a constructive bijection—both composites definitionally or propositionally identity—between the presentation and the concrete source's currents together with the full event family at each current [8]; (ii) an emitter issuing the primitive event at every current c; and (iii) a proof that compilation commutes with emission. Nowhere does R contain a field forbidding magic or external supplements.

Definition 2.1 (Temporal reachability and lawful world states). A **temporal reachability witness** (`SourceNativeTemporalReachableAt`) of a current c on an authoritative root R has exactly two inductive constructors:

1. **finite (`finite`)**: starting from the initial visit of R, finitely many steps advance along the compiler's successor equation $\mathsf{next}_R(c_i) = \mathrm{some}\, c_{i+1}$ to reach c;
2. **post-cofinal (`postCofinal`)**: starting from a source-generated cofinal landing visit, finitely many steps advance along the same successor equation to reach c.

A **lawful world state** of R is a dependent pair $w = (c, h)$, where h is a temporal reachability witness of c; the type of all such states is written $\mathsf{Law}(R)$. The **registered occurrence** (`registeredOccurrence`) of w is the dependent pair

$$\mathsf{occ}(w) = \langle w,\ \varepsilon_R(w) \rangle \in \sum_{u : \mathsf{Law}(R)} E_R(u), \tag{2}$$

where $\varepsilon_R(w)$ is the exact temporal causal event generated by R at w (via its sole constructor `.canonical`). The same visit simultaneously reads off the authoritative temporal evolution $\mathsf{auth}(w)$ and one of five exhaustive primitive dispositions from the same compiler image: native write, relation write, continued transport, Borromean redirect (multi-row coupled-constraint redirect [6]), or local faithful terminal.

The admission surface of the zero law is $\mathsf{Law}(R)$ itself: there is no secondary registry and no caller-submitted world record. Because a post-cofinal history can begin only at a source-generated cofinal landing, no finite visit can masquerade as a cofinal one; because the finite embedding is

injective, reaching the same current label c along distinct causal histories $h \neq h'$ yields distinct lawful world states in $\mathsf{Law}(R)$.

Lemma 2.2 (Root-read mother identity and injective occurrence admission). Inside the zero-axiom kernel (`#print axioms = []`), for every source-native authoritative root R and any lawful world states $w, w' \in \mathsf{Law}(R)$:

1. **Root-read identity ($\mathsf{RootRead}(o) = o$)**: on complete temporal causal root occurrences, the root-coface readout $\mathsf{rootCofaceRead}(o) = o$ is the literal identity map and is consumer-safe with empty `HiddenNullWitness` for **every** downstream consumer (`rootCofaceRead_consumerSafe`, `no_hiddenNull_at_rootCofaceRead`, dual to Theorem 3.3); at an exact temporal visit w, the canonical generated evolution token $\mathsf{generatedAt}(w)$ has its event, predecessor patch stream, current finite patch, and whole-ledger write-back definitionally equal to the root compiler's direct outputs at w (`occurrence_eq`, `priorPatches_eq`, `currentPatch_eq`, `wholeLedgerWriteBack_eq`), and every installed semantic face ϕ restricts directly from that same occurrence via $\mathsf{rootRead}(\phi)$, leaving no second root-level result object outside o.
2. **Injective occurrence admission**: if $\mathsf{occ}(w) = \mathsf{occ}(w')$, then $w = w'$; equivalently, $w \neq w'$ implies $\mathsf{occ}(w) \neq \mathsf{occ}(w')$.

Proof. (1) `rootCofaceRead` returns `o` definitionally on complete temporal root events, so `congrArg consumer` proves universal consumer safety; for fixed (R, w), `SourceNativeTemporalVisitGeneratedEvolutionAt` has the sole constructor `.canonical`, unfolding predecessor patches, current patch, and whole-ledger write-back to the exact root compiler images (`rfl` / `HEq.rfl`), while `SourceNativeRootSemanticFaceAt.rootRead` evaluates directly on `emitted(w.current)`. (2) Taking the first projection pr_1 of the Σ-type equality in (2) gives $w = w'$ in one step; the contrapositive follows immediately. □

Lemma 2.2(2) takes a single line precisely because Definition 2.1 paid the full structural price prior to entry: the type $\mathsf{Law}(R)$ admits only source-generated visits, leaving no unregistered compartment where an off-ledger bit could hide.

2.2 The Boolean Ghost Counterexample and the Single-Root Compile-Time Firewall

We now test this admission gate with a paired positive and negative control (**OB2**). In the fixed verification source package (source Y in Appendix D), let $R_\circ$ be the concrete inquiry-world root, v_1 its initial visit, v_2 the one-step successor visit of v_1, and $w_i = \mathrm{finite}(v_i)$ $(i = 1, 2)$.

Proposition 2.3 (One extra raw bit creates no new world, and the single-root firewall). Inside the zero-axiom kernel (`#print axioms = []`), all four statements hold:

1. **Genuine causal steps carry distinct occurrence identities**: $v_1 \neq v_2$, hence $w_1 \neq w_2$ and $\mathsf{occ}(w_1) \neq \mathsf{occ}(w_2)$.
2. **Unregistered ghost bits collapse at admission**: on $G = \mathsf{Law}(R_\circ) \times \mathrm{Bool}$, set $g_0 = (w_1, \mathrm{false})$ and $g_1 = (w_1, \mathrm{true})$. As raw pairs $g_0 \neq g_1$; yet under the sole lawful admission projection (`.1`),

$$g_0.1 = g_1.1 = w_1, \qquad \mathsf{occ}(g_0.1) = \mathsf{occ}(g_1.1), \qquad \mathsf{auth}(g_0.1) = \mathsf{auth}(g_1.1). \tag{3}$$

3. **Constructors reject ghost arguments and unadmitted raw roots**: passing an extra `Bool` argument to the lawful-state constructor, or attempting to use a raw mixed terminal/write root lacking complete event-inventory admission as a world root, fails type-checking directly (`fail_if_success`).
4. **Single-root compile-time authority firewall**: the active public authority facade (`LivingLawSingleRootAuthorityFacadeRegression`) enforces compile-time `#guard_msgs` guards blocking eight world-altering overreach vectors—(i) pruning the complete actual-event inventory, (ii) borrowing authority from sibling fields or ledger rows, (iii) promoting a completed future oracle to a current atom, (iv) executing inquiry or revision payloads outside the fixed root occurrence, (v) reintroducing parallel lifecycle or revision-selection oracles (`U8RevisionCalculus`, `GeneratedRevisionFamilyAt`, `RevisionSelectionCalculus`), (vi) activating uninstalled public faces, (vii) replacing an inquiry macro-successor with an old-root micro-successor, and (viii) preloading completed tables of future query activations.

Proof. (1) The histories of v_1 and v_2 terminate in distinct inductive constructors (initial vs. step), giving $v_1 \neq v_2$; injectivity of `finite` yields $w_1 \neq w_2$, and Lemma 2.2(2) yields $\mathsf{occ}(w_1) \neq \mathsf{occ}(w_2)$. (2) If $g_0 = g_1$, projecting the second coordinate yields false = true, a contradiction; all three equalities in (3) hold definitionally (`rfl`) since $g_0.1 \equiv w_1 \equiv g_1.1$. (3) and (4) are enforced at type-checking time by constructor signatures, complete-inventory bijection fields, and public import-surface isolation, verified mechanically via `fail_if_success` and `#guard_msgs`. □

Proposition 2.3 cleanly separates two true facts (Figure 2): G is a legitimate mathematical product in ambient type theory with $g_0 \neq g_1$, and after admission there is only a single world state w_1. For the extra Boolean to become a genuine world difference, it must enter a source or root transition—or trigger a subsequent residual or revision event—thereby arriving with a distinct occ just like w_2 in item (1); if it leaves every source-fixed evolution and readout unchanged forever, it is merely a redundant presentation of the same reality.

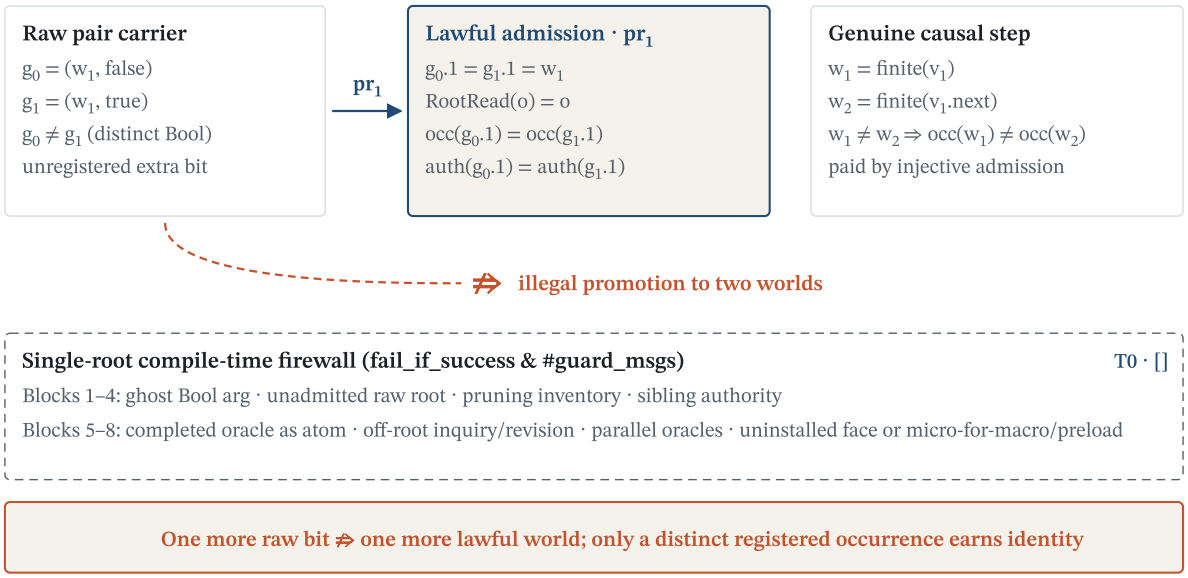


Figure 2 | Ghost-bit interception and single-root firewall. Top left: as raw pairs, $g_0 = (w_1, \text{false})$ and $g_1 = (w_1, \text{true})$ differ in their second coordinate. Top centre: the sole lawful admission projection pr_1 sends both to w_1, obeying $\mathsf{RootRead}(o) = o$ with identical registered occurrences and authoritative evolutions. Top right: a genuine causal difference $w_1 \neq w_2$ arises from distinct root visits and receives distinct occurrences by injective admission. The dashed vermilion curve marks the blocked promotion, while the lower dashed box summarizes the eight overreach entry points blocked at compile time via `fail_if_success` and `#guard_msgs` in the T0 kernel.

2.3 Weak Difference Vocabulary and Non-Circular Zero-Axiom Exclusion of Magic

With injective admission in hand, we prove that **excluding external supplements and magic bits requires zero prohibitory axiom fields, following constructively from positive difference classification alone (OB1, OB3).**

Definition 2.4 (Weak difference semantics, internal determination, external supplement, and magic bit). A **weak difference semantics** $\mathcal{R}$ consists of a carrier type D equipped with five independent predicates: actuality Act, stable reference Ref, real difference Real, redundant presentation Red, and positive structural identity Id. For $d \in D$, define:

- **actual difference** $\mathsf{AD}(d) :\Longleftrightarrow \mathsf{Act}(d) \wedge \mathsf{Ref}(d) \wedge \mathsf{Real}(d)$;
- **internal determination** $\mathsf{ID}(d) :\Longleftrightarrow \mathsf{AD}(d) \wedge \mathsf{Id}(d)$ (an actual difference paired with its positive structural identity witness);
- **external supplement** $\mathsf{ES}(d) :\Longleftrightarrow \mathsf{AD}(d) \wedge \neg\mathsf{Red}(d) \wedge \neg\mathsf{ID}(d)$;
- **magic bit** $\mathsf{MB}(d) :\Longleftrightarrow \mathsf{AD}(d) \wedge \neg\mathsf{Red}(d) \wedge \neg\mathsf{ID}(d)$.

The semantics $\mathcal{R}$ satisfies **positive total reality** (`TotalRealityAt`), written $\mathsf{Total}(\mathcal{R})$, if there is a positive location map `locate` such that

$$\forall d \in D, \qquad \mathsf{AD}(d) \implies \mathsf{Red}(d) \vee \mathsf{Id}(d). \tag{4}$$

Two design choices in Definition 2.4 eliminate circularity. First, $\mathsf{ID}(d)$ is not defined negatively as $\neg\mathsf{MagicBitAt}$; it demands an explicit positive witness of $\mathsf{Id}(d)$, and $\mathsf{Total}(\mathcal{R})$ in (4) states positively

where actual differences land without mentioning magic or supplements. Second, that $\mathsf{MB}(d)$ and $\mathsf{ES}(d)$ share the same three fields is not a duplicate definition, but exposes an immediate logical obligation (`MagicBitAt.toExternalSupplement`): **to claim that a difference is actual and non-redundant while refusing every internal structural determination is ipso facto to demand an external patch written in from outside the theory.**

Theorem 2.5 (Positive total reality constructively excludes external supplements and magic bits). Inside the zero-axiom kernel (`#print axioms = []`), if $\mathsf{Total}(\mathcal{R})$ holds, then for every $d \in D$ both $\mathsf{ES}(d)$ and $\mathsf{MB}(d)$ are empty (imply False).

Proof. Given $e : \mathsf{ES}(d)$, apply `locate` in (4) to $e.\mathsf{actualDifference} : \mathsf{AD}(d)$. Landing in $\mathsf{Red}(d)$ contradicts $e.\mathsf{nonredundant}$; landing in $\mathsf{Id}(d)$ pairs with $e.\mathsf{actualDifference}$ to inhabit $\mathsf{ID}(d)$, contradicting $e.\mathsf{notInternallyDetermined}$. Any $m : \mathsf{MB}(d)$ maps field-by-field via `toExternalSupplement` to $\mathsf{ES}(d)$ and meets the same contradiction. Only constructive conjunction and disjunction elimination are used—neither excluded middle, choice, nor quotient soundness enters. □

Theorem 2.5 is an abstract logical engine: if a caller merely supplied $\mathsf{Total}(\mathcal{R})$ as an assumption, nothing concrete would yet be earned. Theorems 2.6 and 2.8 prove that every authoritative root and every complete living process **generates** Total **unconditionally from its own admission structure**.

2.4 Root-Level Total Reality and Living-Process Total Reality Across Handoffs and Revisions

Theorem 2.6 (Total reality of an arbitrary source-native authoritative root, `RootTotalReality`). Inside the zero-axiom kernel (`#print axioms = []`), for every relation network N, vocabulary V, and source-native authoritative root R, define the root difference semantics $\mathcal{R}_R$:

$$\begin{aligned} D &= \mathsf{Law}(R) \times \mathsf{Law}(R), \\ \mathsf{Act}(w,w') &: \text{ both } \mathsf{auth}(w) \text{ and } \mathsf{auth}(w') \text{ possess canonical visit-compiler receipts,} \\ \mathsf{Ref}(w,w') &: \text{ both sides possess exact temporal root events equal to } \mathsf{occ}(w), \mathsf{occ}(w'), \\ \mathsf{Real}(w,w') &: w \neq w', \qquad \mathsf{Red}(w,w') : \mathsf{occ}(w) = \mathsf{occ}(w'), \qquad \mathsf{Id}(w,w') : \mathsf{occ}(w) \neq \mathsf{occ}(w'). \end{aligned} \tag{5}$$

Then: (a) every pair of unequal lawful states $w \neq w'$ generates its full actual-difference witness $\mathsf{AD}(w,w')$ from R itself; (b) $\mathsf{Total}(\mathcal{R}_R)$ holds; and (c) for every $d \in D$, $\mathsf{ES}(d)$ and $\mathsf{MB}(d)$ are empty.

Proof. (a) For every $u \in \mathsf{Law}(R)$, the root closure supplies the proof that its generated evolution is the canonical visit-compiler image, giving $\mathsf{Act}(w,w')$; $\langle \mathsf{occ}(w), \mathsf{rfl} \rangle$ and $\langle \mathsf{occ}(w'), \mathsf{rfl} \rangle$ give $\mathsf{Ref}(w,w')$; together with $w \neq w'$ they inhabit $\mathsf{AD}(w,w')$. (b) Applying Lemma 2.2(2) to $w \neq w'$ directly constructs the right disjunct $\mathsf{occ}(w) \neq \mathsf{occ}(w')$ of (4). (c) Follows from Theorem 2.5. □

Theorem 2.6 covers all finite, cofinal, and post-cofinal visits within one root, and Proposition 2.3(1) certifies non-vacuity. Yet an evolving system need not remain inside a single local root: it may settle a local ledger at a **faithful local terminal** and hand control to a new root, or cross a **theory-revision** boundary into an expanded vocabulary. Lifting the construction to complete living processes (`RootProcessTotalReality`) closes any seam between root epochs.

Definition 2.7 (Complete living source process). A **living current** over N (collection $\mathrm{Cur}(N)$) is a dependent triple (V, ρ, v) consisting of a vocabulary V, a living root closure ρ carrying its source-fixed terminal-handoff law, and a temporal visit v of ρ. A **complete living source process** $\mathcal{P} = (S, \iota, s_0, \sigma)$ consists of an arbitrary state type S, an **injective registration** $\iota : S \hookrightarrow \mathrm{Cur}(N)$, an initial state $s_0 \in S$, and a one-step successor $\sigma : S \to S$ equipped with a proof that $\iota(\sigma s)$, after erasing the handoff law, equals the next current generated by the root of $\iota(s)$ at that visit, preserving the living root heterogeneously on non-terminal branches and landing on the source-fixed handoff target at local faithful terminals [6].

Theorem 2.8 (Total reality of a complete living process across terminals and revisions, `RootProcessTotalReality`). Inside the zero-axiom kernel (`#print axioms = []`), for every network N and complete living source process $\mathcal{P} = (S, \iota, s_0, \sigma)$, define $\mathcal{R}_{\mathcal{P}}$ on $S \times S$ with Act given by canonical causal "answer-and-next" evolutions at both states, Ref by process-indexed causal occurrences at both states, Real by $s \neq s'$, Red by $\iota(s) = \iota(s')$, and Id by $\iota(s) \neq \iota(s')$. Then every unequal pair generates $\mathsf{AD}(s, s')$ from $\mathcal{P}$, $\mathsf{Total}(\mathcal{R}_{\mathcal{P}})$ holds, and external supplements and magic bits are empty across all root handoffs and revisions.

Proof. $\mathcal{P}$ natively supplies canonical answer-and-next evolutions and emitted occurrences at every state, giving Act and Ref. When $s \neq s'$, injectivity of ι precludes $\iota(s) = \iota(s')$, landing in Id; Theorem 2.5 completes the proof. □

Example 2.9 (A local terminal handoff is not a global halt bit). The fixed verification source package (source Y in Appendix D) constructs an explicit cross-root living process (`local_terminal_handoff_has_structural_identity`): $S = \mathrm{Option}\,\mathbb{N}$, where $\iota(\mathrm{none})$ is the terminal visit of a local faithful terminal root, $\iota(\mathrm{some}\,n)$ is the depth-n visit of a native-write root ($n = 0$ at initial visit), and σ sends $\mathrm{none} \mapsto \mathrm{some}\,0$ and $\mathrm{some}\,n \mapsto \mathrm{some}\,(n+1)$. Because the two vocabularies differ (only the former possesses a faithful terminal fibre) and visit depth recovers n, ι is injective. Across the terminal seam, none and some 0 are distinguished not by an external Boolean "halt flag", but by two distinct complete living currents $\iota(\mathrm{none}) \neq \iota(\mathrm{some}\,0)$.

2.5 Inversion of the Burden of Proof

At this point the T0 kernel has inverted the ontological burden of proof without circularity. "No magic" was assumed nowhere; only **provenance-bearing admission** was defined. This also locates our construction cleanly relative to dependent type theories: we employ dependent Σ-projection [10] to recover lawful states inside an intensional CIC kernel (without adopting the equality-reflection rule of [10] p. 32), and we do not invoke Homotopy Type Theory's Univalence Axiom [11], which governs type equivalence and type equality within a universe; inside a fixed authoritative root or living process, state identity is determined strictly by the single-constructor unfolding tree and registered visits $\mathsf{occ}(w)$ and $\iota(s)$. Any proposed extra bit of reality now faces an exhaustive fork: either it alters a source or root transition (or registers a later residual or revision), thereby entering with a distinct occ or $\iota(s)$ and earning positive structural identity; or it alters no registered occurrence anywhere and collapses to a redundant presentation. The third option—"genuinely world-changing yet permanently off the books"—is constructively uninhabited.

3 Occurred Rigidity, Observer-Quotient Blindness, and the Unstored Future

Once registered differences are proved rigid, two opposite misreadings arise: first, that because one complete current uniquely fixes the next step, any partial observation can reconstruct the entire past; second, that because evolution admits no magic, all concrete events of the infinite future must already sit preloaded in present reality as static inventory. Two sharp results (**OB4, OB5**) separate **past/present occurrence rigidity**, **observer-quotient information loss**, and **on-demand future responsibility**.

3.1 Current Reflection and On-Demand Finite History: Blocking Premise Smuggling

Theorem 3.1 (Current reflection, law-surface invariance, and on-demand finite history). Inside the zero-axiom kernel (`#print axioms = []`), fix any complete living source process $\mathcal{P} = (S, \iota, s_0, \sigma)$:

1. **Same current implies same answer and next**: if $\iota(s) = \iota(s')$, the canonical causal "answer-and-next" evolutions at s and s' are heterogeneously equal (`HEq`), and in particular $\iota(\sigma s) = \iota(\sigma s')$;
2. **Successor divergence reflects into the current**: if $\iota(\sigma s) \neq \iota(\sigma s')$, then $\iota(s) \neq \iota(s')$;
3. **Fixed-process law-surface invariance**: after handoff-law erasure, $\iota(\sigma s)$ equals the next current generated by the root of $\iota(s)$, and both living roots share the exact same law surface (`successor_lawSurface_eq`);
4. **On-demand finite history**: for every reading budget $n \in \mathbb{N}$ and state $s \in S$, σ recursively generates the length-n history $\mathsf{hist}(n, s)$ via $\mathsf{hist}(0, s) = \mathsf{exhausted}(s)$ and $\mathsf{hist}(n+1, s) = \mathsf{step}(\mathsf{hist}(n, \sigma s))$.

Proof. (1) Injectivity of ι gives $s = s'$, making both evolutions identical. (2) Contrapositive of (1). (3) Project the law surface across the first component of the generated-successor proof and apply root law-surface preservation. (4) Structural recursion on n. □

Items (1)–(2) close a subtle formal back door: **assuming an outcome-deciding hypothesis H in a theorem premise and then passing off "provable under H" as "already occurred in this world".** Conditional reasoning remains completely free, producing conditional propositions; yet within a fixed process $\mathcal{P}$, nothing can alter tomorrow's canonical answer or $\sigma(s)$ without already altering the registered living current $\iota(s)$ today (or switching to a distinct source process). Item (3) shows that even crossing a local terminal (as in Example 2.9 advancing $n = 2$ steps from none through some 0 to some 1) cannot smuggle in a theory revision, which requires the independent construction of §6; in item (4), $\mathsf{exhausted}$ signifies only exhaustion of the finite reading fuel n, as the history type contains no "world-stopped" constructor.

3.2 Unseen Is Not Absent: Observer-Quotient Refutation and Global Integral Torsion Blindness

Rigidity belongs to the complete living current; concrete instruments, interfaces, and scalar projections are observation maps $q : A \to O$ into coarser codomains that routinely identify distinct states (**OB5**).

Definition 3.2 (Observer safety relative to a readout and hidden-difference witness). Let $q : A \to O$ be an observation map and $c : A \to K$ a downstream readout map (formally termed a consumer). Call q **safe for** c (`ConsumerSafe`) if $\forall x, y \in A,\ q(x) = q(y) \Rightarrow c(x) = c(y)$. A **hidden-difference witness** (`HiddenNullWitness`) for q relative to c is a tuple $(x, y, p_{\text{obs}}, p_{\text{diff}})$ satisfying $q(x) = q(y)$ and $c(x) \neq c(y)$.

Theorem 3.3 (Refutation of safety certification, full-trace redirect, and complex-observation annihilation).

1. **Irrevocable refutation under post-processing**: given a hidden-difference witness, $\mathsf{ConsumerSafe}(q, c)$ is empty, and every further compression $h \circ q$ ($h : O \to O'$) is refuted by the same pair (x, y).
2. **Full-trace redirect for invisible-scope obligations**: in a native responsibility process, admitting a commitment whose scope is invisible to a scoped observer leaves that observer's filtered list unchanged while strictly increasing the length of the complete current-responsibility observer's list; packaging the admission event with both proofs constructs a provenance-preserving `FullTraceRedirect`.
3. **Annihilation of a global integral torsion class across all finite complexified observations**: in the globally compatible integral cokernel $G = \{(g_n) \in \prod_n C_n : \rho_n(g_{n+1}) = g_n\}$ generated by the canonical arithmetic source, there exists a nonzero torsion class $z \in G$ detected by the stage-one integral readout $J(z) \neq J(0)$ while **every** finite-stage complexified observation in $F(g) = (1 \otimes_{\mathbb{Z}} \pi_n(g))_{n \in \mathbb{N}}$ vanishes identically, $F(z) = F(0)$ [8]. Consequently, **no function** h **whatsoever** (linear or nonlinear) satisfies $J = h \circ F$.

Proof. (1) Substituting $q(x) = q(y)$ into safety contradicts $c(x) \neq c(y)$; applying h preserves equality on (x, y). (2) Scoped filtering drops the newly admitted invisible item whereas slot-by-slot full observation appends a live slot, changing list length. (3) By Theorems 13.1–13.3 of Whole-Ledger Accounting [8] (Appendix A.5): endpoint compatibility places $z \in G$, the unit arithmetic root yields $J(z) = z_1 \neq 0$, and stagewise tensoring of finite-order integral torsion over $\mathbb{C}$ gives $1 \otimes_{\mathbb{Z}} z_n = 0$ without exchanging inverse limits and tensor products, so $F(z) = 0 = F(0)$, ruling out any h with $J = h \circ F$. □

Theorem 3.3 enforces a sharp methodological trichotomy whenever an interface reports "no change": the observation is either proved safe for the target readout, refuted by a hidden-difference witness, or unclassified—never silently equated with ontological absence.

3.3 Tomorrow Is Not Preloaded Inventory; It Is Generative Responsibility

Taken together, Theorems 3.1 and 3.3 dispel deterministic fatalism. What present reality owns is the **current state paired with its one-step generative law** (s_t, σ), not an infinite completed tuple $(s_t, s_{t+1}, s_{t+2}, \dots)$. Theorem 3.1(4) provides on-demand prefix unfolding for any finite fuel n, storing no completed future table; likewise, the all-future model of §5 characterizes equivalence across finite action words without materializing unexecuted future events. In state-machine refinement theory [16], future-dependent *prophecy variables* serve as proof-level auxiliaries augmenting a lower-level specification while projecting back onto the same visible behaviours; in our ontological state admission, by contrast, the lawful current state itself stores no unexecuted

future-dependent data—preloading ungenerated future outcomes directly as present atomic state would introduce an unaccounted bypass oracle (blocked at compile time by items (iii) and (viii) of Proposition 2.3(4)). **Past and present are rigid because they are registered; tomorrow remains open because its generative bill has not yet been paid.**

4 Universal Three-Domain Formation from the Fixed Spin-Pair Mother Source $S_\star$

Sections 2 and 3 established internal rigidity for each given authoritative root R and process $\mathcal{P}$. A deeper ontological question immediately follows: **where do all those diverse roots, inquiries, and processes come from?** If every physical domain or inquiry language required choosing a separate underlying source from outside, "which source was chosen" would itself become an unaccounted degree of freedom.

At the standard three-axiom analytic layer (T1), we prove the **Universal Single-Mother Formation Theorem** (**OB6**–**OB7**): without object-by-object source selection, the **single fixed physical spin-pair mother root** $S_\star$—canonically generated from primitive event structure—universally forms every lawful world, every complete inquiry, and every macro-process on the `Type 0` admission interface via three stages: *finite-programme approximation*, *pointwise uniform completion*, and *canonical realization with left-inverse restriction*.

4.1 The Endogenous Mother $S_\star$, Finite Programmes, and Pointwise Uniform Completion

Fix the physical mother root $S_\star$, the authoritative ledger root of the two-state `SpinPair` living root in the continuous–quantum unified field flagship [7]. Let us address at once a natural question: **does fixing $S_\star$ amount to hand-picking a privileged source or tuning parameters?** No: Theorem 9.3 (second edition Theorem 8.3) of the physics flagship [7] proves that primitive event structure endogenously generates a canonical root—without selecting from a candidate pool—whose n-th visit reproduces for every $n \in \mathbb{N}$ the exact current of the n-th visit of the original mother history. Here we prove that this single minimal physical root suffices to form the entire `Type 0` admission forest.

Let $\mathbb{S} = \mathbb{R}^{\mathbb{N}}$ denote real sequences. At material rank r, let X_r be the input carrier and $\mathcal{A}_r$ a family of point-separating observation addresses. A **finite mother programme** at rank r pairs a finite address tuple $(a_0, \dots, a_{n-1}) \in \mathcal{A}_r^n$ with one lawful visit $\nu \in \mathsf{Visit}(S_\star)$; evaluation zero-pads the n observed real values into a stream and feeds it into the finite evaluator of $S_\star$:

$$\begin{aligned} \mathsf{Prog}_r &= \coprod_{n\in\mathbb{N}} \mathcal{A}_r^n \times \mathsf{Visit}(S_\star), \\ L_r((a_0,\dots,a_{n-1};\nu),\, x) &= \mathsf{finiteLaw}_{S_\star}\big(\nu,\ \mathrm{pad}_n(\mathrm{obs}_r(a_i,x))_{i<n}\big), \\ \mathcal{R}_r &= \mathrm{im}\, L_r \subseteq (X_r \to \mathbb{S}), \qquad \mathcal{M}_r = \widehat{\mathcal{R}_r}. \end{aligned} \tag{6}$$

Here $\mathcal{M}_r = \widehat{\mathcal{R}_r}$ is the uniform completion with respect to the product uniform structure (input by input, finite output prefix by prefix), with canonical dense uniform embedding $j_r : \mathcal{R}_r \hookrightarrow \mathcal{M}_r$.

Both low-rank and high-rank materials follow (6), with high-rank input carriers X_r taking already formed low-rank complete material spaces.

Definition 4.1 (Formation by $S_\star$). A material $m \in \mathcal{M}_r$ is **formed by** $S_\star$ if m belongs to the topological closure in $\mathcal{M}_r$ of the finite-programme evaluation image $\{j_r(L_r(p)) : p \in \mathsf{Prog}_r\}$.

Observe the logical status of Definition 4.1: **being formed by $S_\star$ is a theorem proved unconditionally inside the construction, not an extra hypothesis imposed on admitted objects.** Because the finite-programme image equals $\mathcal{R}_r$, which is dense in its own uniform completion $\mathcal{M}_r$, every $m \in \mathcal{M}_r$ satisfies Definition 4.1 automatically. The substantive mathematical work lies in constructing internal ranks and separating addresses for any given target world, inquiry, or macro-process—without adding representability premises to its original type—so that canonical realization reconstructs the object and recovers all its operations by left-inverse restriction.

4.2 Universal Three-Domain Formation and Exclusion of an Independent Second Base Source

Theorem 4.2 (Universal three-domain formation on the `Type 0` interface from one mother $S_\star$ [7]). Fix $S_\star$. On the original admission interface (objects in the lowest universe `Type 0`), with zero added representability or coverage hypotheses, all three universal statements hold simultaneously:

1. **All lawful worlds ($\forall N, V, R, w$).** For every relation network N, vocabulary V, source-native authoritative root R, and lawful world state $w \in \mathsf{Law}(R)$, there exist an internally constructed rank r and a complete material $m \in \mathcal{M}_r$ formed by $S_\star$ such that the canonical state realization map produces some(V, R, w) from m, reads off the complete world (N, V, R, w) with an explicit left inverse to material encoding, recovers registered occurrence, authoritative evolution, positive disposition, prior and current patches, whole-ledger write-back, next current, and step advance up to heterogeneous equality, and endows the recovered root with total reality $\mathsf{Total}(\mathcal{R}_R)$.
2. **All complete inquiries ($\forall N, V, q$).** For every N, V and complete inquiry state q, there exists a high-rank material formed by $S_\star$ whose read-off inquiry world equals $(N, V, \mathrm{create}(q))$, whose formed clauses equal the answer, action, and revision clauses of q, whose clause compilation commutes heterogeneously with original query compilation at every query and exact causal event, and whose parent low-rank material spaces are recovered by left-inverse restriction.
3. **All macro-processes ($\forall P$).** For every native inquiry-engine macro-process P, there exist a high-rank material m formed by $S_\star$ and a formed process P_m satisfying

$$\mathsf{form}(m) = \mathrm{some}(P_m), \qquad \mathsf{restrict}(P_m) = P, \tag{7}$$

retaining all parent low-rank material spaces by left-inverse restriction (`ParentsRetained`), and recovering state carrier, per-state query nodes, initial state, successor, commuting query compilations, existing engines, activations, and complete activation laws; moreover, for every execution trajectory $\mathcal{T}$ and depth $n \in \mathbb{N}$, under the type transport T_m induced by (7),

$$\widetilde{\mathcal{T}}.\mathsf{stateAt}(n) = T_m(\mathcal{T}.\mathsf{stateAt}(n)), \quad \widetilde{\mathcal{T}}.\mathsf{tickAt}(n) \doteq \mathcal{T}.\mathsf{tickAt}(n), \quad \widetilde{\mathcal{T}}.\mathsf{tickAt}(n).\mathsf{next} = T_m(\mathcal{T}.\mathsf{tickAt}(n).\mathsf{next}). \tag{8}$$

Load-bearing proof architecture. The central obstacle is bridging finite observation to full functional domains: each programme $(a_0, \dots, a_{n-1}; \nu)$ inspects only finitely many coordinates, whereas target state and query maps act on full carriers $f : X_r \to \mathbb{S}$. Three steps close the gap (Appendix B.1–B.3): First (**dense approximation and bijective completed readout**): for any finite subset $F \subset X_r$, output prefix $b \in \mathbb{N}$, and $\varepsilon > 0$, picking one separating address per distinct pair in F yields an address tuple injective on F; the finite-evaluator approximation theorem of $S_\star$ (evaluated at mother-history visit $10 + c$) then matches f within ε on $F \times \{0, \dots, b-1\}$. Because (F, b, ε) parametrizes the basic entourages of the pointwise uniform structure, $\mathcal{R}_r$ is dense in $X_r \to \mathbb{S}$, and the completed uniform embedding $\mathsf{read}_r : \mathcal{M}_r \to (X_r \to \mathbb{S})$ has closed dense image, hence is bijective. Second (**unified disjoint-sum witness and left-inverse restriction**): for any macro-process P, assembling states and state-indexed queries into $\mathcal{U}_P = \mathrm{State}(P) \sqcup \coprod_s \mathrm{Query}(P.\mathsf{stateAt}(s))$ inside `Type 0` internally supplies a sufficient rank and separating addresses; interleaving stream pairings merge state, node, address, and successor materials into a single joint witness m whose stream projections provide exact left-inverse parent restrictions. Third (**canonical realization and all-depth trajectory transport**): reconstructing the state-relabelled process P_m from m and restricting along the generated state equivalence yields the exact record identity (7), which transports activation laws and discrete steps across all depths $n \in \mathbb{N}$ in (8) using the same fixed material m. □

Three ontological and methodological corollaries stand out immediately:

- **No independent second base source on `Type 0`**: item (1) quantifies universally over **all** authoritative roots ($\forall R$). Any root R' proposed on the `Type 0` interface as an "independent second foundational source", together with all its lawful world states, is already formed and recovered field-for-field from completed finite programmes of $S_\star$.
- **No operational-probabilistic primitives, modal task postulates, or specification auxiliaries**: unlike operational quantum derivations working in an operational-probabilistic framework with purification [14], constructor theory formulating laws via possible/impossible tasks and the interoperability principle (Principle III) [15], or refinement completeness augmenting lower-level specifications under regularity conditions with auxiliary history and prophecy variables to construct refinement mappings [16], Theorem 4.2 synthesizes separating addresses internally on `Type 0` disjoint sums and recovers full histories by strict left-inverse restriction from the single deterministic physical mother $S_\star$.
- **Formation completion is not preloaded runtime inventory**: $\mathcal{M}_r$ in (6) is static formation data fixed by a universal property; Theorem 4.2(3) recovers the **successor law** and finite-depth histories (8), while concrete subsequent occurrences are still generated dynamically step by step (§3.3).

4.3 Direct Instantiation in the Unified Field and the Dirac-Swap Negative Control

A universal theorem proves its mettle when put directly to work on a full-scale physical unification (**OB7**).

Corollary 4.3 (Five non-bypass downstream verification chains [7]). Five audited theorems invoke the single declaration `root_admission_fixed_mother_complete_realization` of

Theorem 4.2 directly—verified by syntax-tree inspection to **bypass none of the three low-level coverage lemmas or four low-level recovery lemmas**, with full dependency closure restricted to the standard three axioms: (1) universal world formation and whole-ledger/next-current recovery; (2) universal inquiry formation, parent retention, and per-event compilation recovery; (3) universal macro-process identity (7); (4) recovery of the physical unified-field inquiry seal, next engine, and complete activation law at evolution state 16; and (5) simultaneous recovery, for **every** depth $n \in \mathbb{N}$, of step $16 \to 17$, the n-th state and step, mother-history physical visit 10, and `SameOccurrenceActivation`.

Theorem 4.4 (Continuum nine-field configuration, $4\sqrt{2}$ quantum pairing, and macro-step successor on the same `SpinPair` root [7]). From the `SpinPair` living root, an endogenously generated faithful realization $\mathcal{Q}$ satisfies: (i) two visits form an internally determined difference iff their representation visits in $\mathcal{Q}$ differ; (ii) any other faithful realization admits a unique commuting presentation; (iii) every event recovers all original projections, writers, and whole-ledger write-backs; and (iv) at every event of the physical visit, the read-off configuration passes four-dimensional $\mathrm{Spin}^{+}(1,3) \times \mathrm{SU}(7)$ nine-field classical world admission and 252-complex-dimensional chiral matter recovery, the quantum readout equals the step answer a, and at every spacetime point x and **every** complex-linear endomorphism $A \in \mathrm{End}_{\mathbb{C}}(\mathbb{C}^{252})$, the exact finite-$\hbar$ pairing holds:

$$\chi_x(A\,\psi_x) = 4k\,\langle a_x,\ R(A)\,a_x\rangle, \qquad k = \sqrt{2}, \quad R(A) = SPAE, \tag{9}$$

while erasing the step's next node recovers the exact next current compiled at that event.

Equation (9) locks §2's internal determination, §4's single-mother formation, and continuum–quantum physics onto one event: the factor $4\sqrt{2}$ is the parameter-free product of classical amplitude 2 ($\psi = 2Ev$) and independent dual scale $2\sqrt{2}$, while $R(A) = SPAE$ compresses A via isometric occupation embedding E, projection $P = E^*$, and the dual-induced Dirac two-spinor block swap S. As shown by the paired negative control in [7] (Example 6.6), omitting S leaves the initial ($\theta = 0$) co-axial gauge current at $+1/2$ by coincidence, yet plunges the dark-moment ($\theta = \pi/2$) current from $+1/2$ to $-1/2$—an error of 1. While the field equations and 504-mode operator algebra are paid by the constructions of [7], Theorem 4.4 certifies that the single mother $S_\star$ hosts and recovers them without loss.

5 Endogenous Residual Requests, Well-Founded Payment, and Paid-AST Homological Continuation

Sections 2–4 settled provenance standing and universal formation. Yet a skeptic might still suspect that a ledger merely stamps receipts after external machinery finishes computing. **A genuine source-native ledger must generate the next computational request endogenously from its own unpaid semantic residual, pay that exact debt out of its own syntax budget, and—in the new v2 continuation loop (OB12–OB13)—recover the paid AST to drive ongoing homomorphic methods and cochain differentials.**

5.1 Nonlinear Source Expressions, Node-Wise Syntax Cost, and Residual Requests

Let S be an arbitrary type of sorts, each $s \in S$ carrying an abelian group V_s and variable type X_s; an environment Γ assigns group values to variables. Source expression trees (ASTs) are built inductively from variables, constants, addition, unary group-homomorphism actions, and bilinear (biadditive) actions. Writing $\mathrm{ev}_\Gamma(a)$ for evaluation under Γ and $D_a(\Gamma, \delta)$ for the recursively defined update increment (effect), structural induction yields the exact finite-difference identity [8]:

$$\mathrm{ev}_{\Gamma+\delta}(a) = \mathrm{ev}_\Gamma(a) + D_a(\Gamma, \delta). \tag{10}$$

At bilinear nodes $B(u, v)$, $D_{B(u,v)}(\Gamma, \delta)$ retains left, right, and nonlinear cross increments $B(du, dv)$, never misidentifying nonlinear effects with evaluation on δ alone (sharing the exact abelian finite-difference expansion of static λ-calculus differentiation [18] Theorem 3.11 and DBSP stream calculus [19] on $\mathbb{Z}$-sets in §4.1 and Theorem 3.4, and extending it below to node-wise syntax debit, paid-AST recovery, and cochain differentials).

Each AST a carries a natural-number **syntax cost** $c(a) \in \mathbb{N}$: variable 1, constant 0, unary action $c(\mathrm{sub}) + 1$, addition and bilinear action $c(\mathrm{left}) + c(\mathrm{right}) + 1$. Every primitive reduction step $a \rightsquigarrow a'$ preserves evaluation and strictly decrements cost, $c(a) = c(a') + 1$; a complete execution state $\sigma = (p, \tau)$ pairs the current reduced expression p with the ordered paid step trace τ from a, obeying $|\tau| + c(p) = c(a)$.

At an admitted root occurrence o with whole ledger $L(o)$ and non-refilling write-back E_o, an **original residual material** $m = (\Gamma, \delta, a, \sigma = (p, \tau), e)$ $(e \in L(o))$ defines the **literal residual AST** r_m and **next endogenous request** Q_m:

$$r_m = a + (-\mathrm{id})(p), \qquad Q_m = (\Gamma + \delta,\ r_m,\ e), \qquad c(r_m) = c(a) + c(p) + 2. \tag{11}$$

As hand-checked in Example 1.2(b) for $a = x \cdot x$, $\Gamma(x) = 2$, $\delta(x) = 2$, and $p = \mathrm{const}\, 4$, Equations (10)–(11) give residual value $\mathrm{ev}_{\Gamma+\delta}(r_m) = 12 = D_a(\Gamma, \delta)$ and syntax budget $c(r_m) = 3 + 0 + 2 = 5$.

5.2 All-Future Equivalence of Residuals and Endogenous Request Birth (OB8)

Theorem 5.1 (All-future residual equivalence and endogenous request generation [8]). Fix any source-generation family $\mathcal{F}$, complete initial frame I, programme cfg, and language identity $\mathsf{cfg.LowVar} = X$. The source run endogenously generates a strictly increasing sequence of admission stops cofinal in discrete evolution steps ($n \in \mathbb{N}$). At stop n, let q_n be the endogenous request generated from the $(n+1)$-th admission packet's root current, and b_n the exact transport of the formal paid boundary $[a] - [p]$. Then for **every** finite allowed action word h and **every** site $j \in \mathbb{N}$,

$$\mathrm{Obs}_j(A_h(\iota_{n+1}[q_n.\mathrm{AST}])) = \mathrm{Obs}_j(A_h(\iota_{n+1} b_n)), \tag{12}$$

where ι_{n+1} is the dependent direct-sum site embedding, A_h iterates allowed source actions letter by letter, and Obs_j reads the full `(old value, effect)` pair at site j. Moreover, the actual next admission current equals the birth programme's canonical successor, q_n is generated directly from

the transported residual material, and both old and born row destinations come from the same whole-ledger write.

Load-bearing proof steps. Inside the free $\mathbb{Z}$-module of formal expression words, let $Z = \{w : \forall\Theta,\ \mathrm{ev}_\Theta(w) = 0\}$ be the **all-environment zero submodule**. Linearity of evaluation gives, for every material m,

$$[r_m] - ([a] - [p]) \in Z. \tag{13}$$

What vanishes across all environments is the difference between the two representations, not r_m itself. For any $w \in Z$, (10) forces both prior and updated evaluations—hence also the effect—to vanish everywhere, giving pair evaluation $(0, 0)$. Because literal pair lifting `liftExpr` evaluates under $\mathrm{pairEnv}(\Gamma, \delta)$ to $(\mathrm{ev}_\Gamma, D(\Gamma, \delta))$ and every pair environment decomposes into its two coordinate projections, `liftExpr` maps Z into the pair-language zero submodule; language and packet transports likewise preserve Z. One site action therefore maps site-k elements of Z into site-$(k+1)$ elements of Z. Induction on the length of h proves that every $\mathrm{Obs}_j \circ A_h$ annihilates (13), establishing (12); identifying $q_n.\mathrm{AST} \equiv r_{m_n}$ completes the proof (Appendix C.1). □

5.3 Strict Same-Debt Debit, Zero Refill, and Suffix Settlement (OB8)

Installing the complete `(old value, effect)` pair readout into the calculation port of $\mathcal{F}$ inside the source (leaving its base reader, writers, stock callbacks, and source seed untouched) yields the derived family $\widehat{\mathcal{F}}$ [8].

Theorem 5.2 (Source effect evaluation and strict same-debt debit). At any admission stop of $\widehat{\mathcal{F}}$ on the current read from original residual material, set $\epsilon = D_a(\Gamma, \delta)$, $Q = c(r_m)$, and let b be the birth object of that stop frame. Then b's registered input equals Q_m, the next canonical request evaluates to ϵ, b has already executed its birth step leaving syntax budget $Q - 1 > 0$, and its actual action necessarily takes the paid reduction branch—producing a same-debt payment π_b whose target uniquely locks onto the original mathematical debt lineage, strictly decrements budget $(Q - 1 \to Q - 2)$ without refill, and agrees heterogeneously with the original row's destination in the whole-ledger successor.

Proof. The stop receipt fixes a constant paid endpoint $p = \mathrm{const}\, v$, and trace soundness gives $v = \mathrm{ev}_\Gamma(a)$, so (10)–(11) yield

$$\mathrm{ev}_{\Gamma+\delta}(r_m) = \mathrm{ev}_\Gamma(a) + D_a(\Gamma, \delta) - v = \epsilon. \tag{14}$$

By (11), $Q = c(a) + c(p) + 2 \geq 2$; since b is read after one birth step, its registered clock leaves $Q - 1 \geq 1 > 0$. Were the actual action a mathematical settlement, settlement would require remaining cost 0, a contradiction; branch elimination therefore extracts a genuine paid step with strict unit debit and monotonic non-refill. Sum-type claim disjointness (original mathematical debt on the left, inherited rows on the right) guarantees unique same-debt targeting, and step induction identifies the whole-ledger destination (Appendix C.2). □

Theorem 5.3 (Suffix completion, receipted answer, and canonical successor delivery). Advancing $Q - 1$ further steps from b's current reaches the exact zero-cost completion state with value ϵ. There the actual action uniquely takes the mathematical settlement branch,

and the target whole-ledger compiler emits a debt-admission action G delivering in one step: the receipted answer, canonical causal "answer-and-next", exact destinations of old and born rows in the shared whole-ledger write, and successor lawfulness, all consumed together with the paid-state and trace-effect identities.

Zero numerical effect still pays syntax cost; new obligations never refill old debt. For $a = \mathrm{const}\, v$ and $p = \mathrm{const}\, v$, $\epsilon = 0$ under every increment, yet $r = \mathrm{const}\, v + (-\mathrm{id})(\mathrm{const}\, v)$ has syntax cost $Q = 2$: the birth step reduces negation ($2 \to 1$) and the next paid step reduces addition ($1 \to 0$) to $\mathrm{const}\, 0$. Zero numerical change erases neither reduction step nor its payment lineage; any subsequent request enters the right sum-type branch paying its own first step rather than refilling prior zero-budget debt. This delineates source-native syntax credit cleanly from linear logic's exponential control of copying and discarding marked formulas [12], amortised separation logic's LP-inferable credit assertions and fallible environmental `acquire` requests (§3.6) [13], and provenance semirings' polynomial homomorphism factorization of positive relational queries [17].

5.4 Continuing Homomorphic Action on Paid ASTs and Primitive-Replay Cochain Differentials (OB12–OB13)

Whereas the first edition stopped at single-request completion, v2 closes the ongoing algebraic and homological loop (**OB12, OB13**): **a newly settled syntax tree is not a dead receipt, but live syntactic fuel driving subsequent homomorphic methods and cochain differentials.**

How can a paid AST be recovered across cofinal stops without an external lookup cache? Inspect the two-layer residual born after settlement: the first residual $e_0 = a + (-\mathrm{id})(p)$ settles to $p_\star = \mathrm{const}\, \epsilon$, which the next birth wraps into $e_0 + (-\mathrm{id})(p_\star)$. Defining the syntactic extractor retainedCode to match the right summand $p_\star$ and apply unary negation ($-\mathrm{id}$) once more yields $(-\mathrm{id})((-\mathrm{id})(p_\star))$. By abelian-group involution $-(-u) = u$, the residual syntax itself restores the exact paid AST!

Theorem 5.4 (Paid-AST recovery, continuing cofinal homomorphic execution, and unified four-face installation, OB12). Let $f : W_s \to W_s$ be any **pre-fixed** additive homomorphism (lifted across pair levels by $f^{(0)} = f$, $f^{(n+1)} = f^{(n)} \times f^{(n)}$):

1. **Universal single-transition recovery and unified face installation**: for **every** sorted abelian-group family W, **every** source-generation family $\mathcal{F}$, and any admission packet pkt_k of the derived method run, retainedCode on the next born packet equals heterogeneously the double-negation image of the prior packet's paid-state AST (`after_native_retained`), evaluating to the prior result v_k (`after_native_value`); evaluating the next packet's linear method node under $\mathrm{pairEnv}(\Gamma, 0)$ yields

$$v_{k+1} \doteq (f^{(n_k+1)}(v_k),\ 0) \qquad (\texttt{after_native_whole_action},\ \texttt{actual_method_step}), \quad (15)$$

   while every event of the complete query write trace (`completeQueryWritten`) enters `writtenStock` through the packet's own callback (`actual_complete_query_written`), and unified **algebraic, combinatorial, topological, and logical** faces (`UnifiedFourFace`, together with homologous **relation** and **cochain** faces) are installed prior to emission with face.rootRead = packet.

2. **Continuing closure across all cofinal stops on the lookahead runtime**: on the lookahead source runtime `sourceRuntime` induced by a complete complex inner-product (Hilbert) space $\mathcal{H}$, living root closure, temporal visit, recognition datum (mapping root occurrences into $\mathcal{H}$), and obstacle-evolution calculus, across every cofinal admission stop $k \in \mathbb{N}$ and every finite update horizon $U \in \mathbb{N}$, homomorphic recursion (15), callback stock containment, and unified face installation iterate across $1 + d_k$ sealed-field steps connecting consecutive stops (`cofinal_paid_method_faces`, `finite_paid_method_faces`).

While Theorem 5.4 propagates paid values across cofinal stops, Theorem 5.5 opens up **every intermediate primitive step** within a word execution to prove that algebraic relations and cochain differentials arise from the very same execution trace (**OB13**).

Theorem 5.5 (Whole-word primitive step replay and stock-boundary / cochain-differential continuation, OB13).

1. **Universal contextual step replay with exact syntax charge**: for **every** relation network N, **every** complete living source process $\mathcal{P}$, sorted abelian-group carrier, environment readout, and nonlinear source expression, `WordExecution` recursively embeds every primitive reduction `Step` into the contextual environment, charging exact remaining substituted syntax cost (`forward_charge`). On the sealed arithmetic–Fock run `sourceRuntime` at any start index `start` with receipt-determined stop stop = start + receipt.1, both the original paid trace and the replay exposure (`nativeReplayAt`) enter `birthStock` and the next complete query via the birth callback (`native_replay_written`, `native_replay_next_query`).
2. **Same-stop coincidence of stock boundary, chain morphism, cochain differential, and physical readback**: at stop, writing $\mu_{\mathrm{stop}} = \mathsf{sourceRelations}(\mathrm{stop})$ for the integral source relation word, $\partial_{\mathrm{act}} = \mathsf{actualBoundary}(\mathrm{stop})$ for the stock-generated closure boundary, $\mathcal{C}^{\bullet} = \mathsf{actualCochain}(\mathrm{stop})$ for the history-generated cochain complex, τ_1 for the substitution chain morphism, E_{stop} for the active pair environment, and $E_{\mathrm{src}} = E_{\mathrm{stop}+1}$ for the pre-substitution source environment, all four differentials coincide:

$$(d^{0,1}_{\mathcal{C}}(\partial_{\mathrm{act}})).\mathrm{val} \;=\; \partial_{\mathrm{act}}.\mathrm{val} \;=\; \mathsf{relMap}_{E_{\mathrm{stop}}}(\tau_1(\mu_{\mathrm{stop}})) \;=\; \mathsf{subst}_{\mathrm{pair}}(\mathsf{relMap}_{E_{\mathrm{src}}}(\mu_{\mathrm{stop}})), \tag{16}$$

with completion projection $\Pi_{\mathrm{comp}}(d^{0,1}_{\mathcal{C}}(\partial_{\mathrm{act}})) = 0$; the unified four-face coimage operator maps the canonical coimage class of the source relation word to that of ∂_{act} compatibly with dual evaluation (`actual_coimage_boundary`, `actual_coimage_differential`), tree erasure recovers the original unfolding while its root second coordinate reads back $\mathsf{face.rootRead}$ (`source_occurrence_erasure`, `source_occurrence_readback`), and at the same stop index stop the action reads back the next-tick physical state and forced trace (`physicalReadback`) while delivering the living-law-preserving whole-ledger successor and (receipt.1+1)-step sealed inquiry segment (`actual_relation_cochain_method`).

Proof sketch (complete derivations in Appendix C.3–C.4). Theorem 5.4 proves by induction along the paid prefix that the registered input expression stays invariant until settlement (`paid_prefix_expression`), extracts the double-negated paid AST at birth, applies $\mathrm{eff}(\mathrm{expr}, \Gamma, 0) = 0$ to obtain (15), and routes the query trace into born stock via `birth_keeps_callback`. Theorem 5.5 inducts over all nine `Step` constructors—never conflating endpoint equality with trace identity—and

chains the telescoping trace-boundary identity, the substitution square `boundary_square`, and `natComplex_d_zero_one` to establish (16), concluding via `map_map` and `map_id` on the single-constructor unfolding tree together with same-stop physical readback and segment splicing. □

Example 5.6 (Homomorphic continuation and three-step homological boundary on the running square example). Plugging $a = x \cdot x$ from Example 1.2(b) into Theorems 5.4 and 5.5 makes both high-order constructions concrete:

- **Homomorphic continuation (OB12)**: after $r_m = x \cdot x + (-\mathrm{id})(\mathrm{const}\, 4)$ settles at $x = 4$ to $p_\star = \mathrm{const}\, 12$ (pair value $(12, 0)$), the next birth registers $r_m + (-\mathrm{id})(\mathrm{const}\, (12, 0))$. Extracting $(-\mathrm{id})((-\mathrm{id})(\mathrm{const}\, (12, 0)))$ via retainedCode restores $(12, 0)$ under zero increment, and applying $f(u) = 3u$ in (15) at the next cofinal stop outputs $((36, 0), 0_{W^2})$ (with pair-group zero $0_{W^2} = (0, 0)$) while logging the query trace in born stock.
- **Stepwise replay and cochain boundary (OB13)**: replaying $x \cdot x$ at $x = 4$ executes three primitive steps—left binding $[x \cdot x] \to [4 \cdot x]$, right binding $[4 \cdot x] \to [4 \cdot 4]$, and bilinear constant reduction $[4 \cdot 4] \to [16]$. Summing their signed stock exposures telescopes to $\partial_{\mathrm{act}} = ([x \cdot x] - [4 \cdot x]) + ([4 \cdot x] - [4 \cdot 4]) + ([4 \cdot 4] - [16]) = [x \cdot x] - [16]$, whose evaluation vanishes $(4^2 - 16 = 0)$, while the three stored intermediate steps distinguish genuine execution of $x \cdot x$ from an uncomputed guess of 16 (Figure 3).

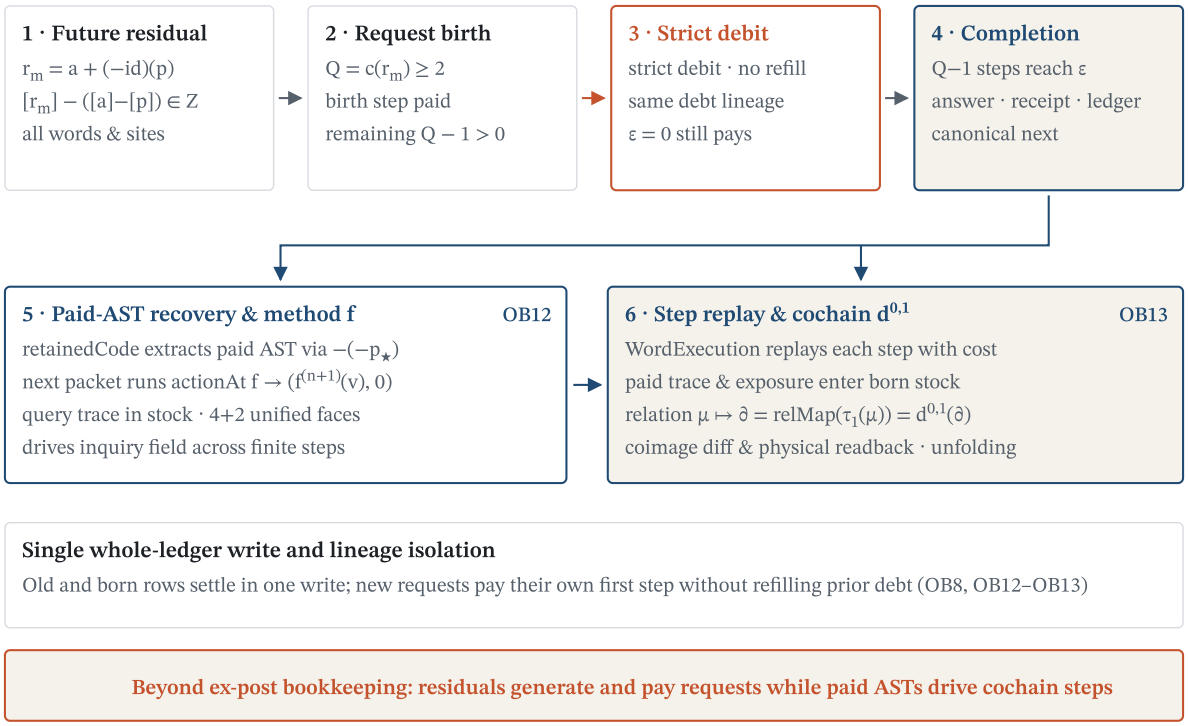


Figure 3 | Residual request, same-debt payment, and paid-AST differential continuation. Top row (OB8): the all-future residual births an endogenous request whose positive syntax budget enforces strict same-debt debit without refill before suffix completion delivers ϵ, the receipted answer, whole-ledger write, and canonical successor. Middle left (OB12): `retainedCode` extracts the paid AST via $-(-p_\star)$ to execute lifted homomorphic method $f^{(n+1)}$ across subsequent cofinal stops, writing complete query traces into born stock and installing unified four-face plus relation/cochain faces. Middle right (OB13): `WordExecution` replays every primitive step with exact cost so that source relations μ update stock boundary ∂, $\mathsf{relMap}(\tau_1\mu)$, $d^{0,1}(\partial)$, and physical readbacks in lockstep. Bottom row: shared whole-ledger write and strict isolation between prior and newly born debt lineages.

6 Rooted Expressibility Bankruptcy, Zero-Axiom Initial Cofaces, and Shared First-Write Revision

When an existing theoretical vocabulary runs aground on a genuine problem, science revises its language. Yet theory revision has historically been the easiest seam through which unaccounted privilege slips in: **could a constitutional revision merely rename an old counterexample while quietly writing off unsettled prior obligations?** Closed entirely inside the **zero-axiom constructive kernel (T0, `#print axioms = []`)**, three theorems (**OB9**) separate *what bankruptcy generates*, *how admissible revisions factorize*, and *how old and new obligations are disposed of together*.

6.1 Rooted Double Bankruptcy and the Zero-Axiom Categorical Initial Coface

A legitimate revision trigger is not an arbitrary wish to switch vocabularies; the revision gate opens only when an existing law suffers **simultaneous emptiness of expression and lawful realization** at an actual root occurrence.

Definition 6.1 (Rooted expressibility and realization double bankruptcy). Fix an old source-native authoritative root R_{old}, a temporal visit v, an old theory state Θ, and an obstruction o at support s. An **actual expressibility failure** of Θ at o consists of a **double-emptiness** proof: both the expression fibre and the lawful-realization fibre of Θ for that demand are empty. The failure is **rooted** at v (`RootedActualExpressibilityFailureAt`, written η) if it further carries coherent same-root proofs: Θ equals the law surface owned by R_{old} at this epoch; the failure payload equals the active reading and canonical token of a projection face installed at v; the source-owned obstruction calculus emits its canonical event on o at support s; and R_{old} carries a unique open demand row e whose theory-audit disposition commutes with whole-ledger compilation, paired with e's causal authority and causal successor at v [6]. Its constructor is closed to external callers and minted solely by the root projection face.

Theorem 6.2 (Obstruction-generated zero-axiom minimal coface, categorical initiality, and unique factorization). Inside the zero-axiom kernel (`#print axioms = []`), without invoking choice, every rooted double bankruptcy η directly generates its minimal coface extension (`root_obstruction_generates_minimal_coface`):

1. **Coordinate extension and old-theorem survival**: over the complete open ledger L_+ at the old successor support, η constructs the disjoint-union carrier and initial budget map

$$C_\eta = L_+ \sqcup \{\star_\eta\}, \qquad \beta(\mathrm{inl}\, e) = \beta(e), \quad \beta(\mathrm{inr}\, \star_\eta) = 0. \tag{17}$$

   The old open ledger embeds via inl, while the constructive retraction maps $\star_\eta$ to the target demand row already carried by η, recovering every old row identically (`backward_forward`) while leaving $\star_\eta$ with no old preimage; incidence and claim coordinates likewise adjoin one preimage-free generator whose claim holds at the new support with a unique η-indexed semantic change, and every old expression theorem survives intact on the embedded old face.
2. **Categorical initiality and unique faithful injection**: for **every** admissible coface revision $\mathcal{A}$ (supplying constructive retractions of old open rows, incidences, and claims together with one preimage-free zero-budget generator and nonincreasing inherited budgets), there exists a canonical faithful coface morphism $F : \mathcal{C}_\eta \to \mathcal{A}$ (`minimal_coface_initial`) whose row, incidence, and claim maps are all **injective**, preserve old embeddings and new generators, and never increase budgets; moreover, inside the class of faithful coface morphisms, F is **pointwise unique** (`minimal_coface_factorization_unique`).

Proof. Disjoint sum constructors preclude any old preimage of $\star_\eta$, and the retraction evaluates at $\star_\eta$ to η's built-in demand row without invoking choice. For any admissible $\mathcal{A}$ with old embedding j and new generator a, set $F(\mathrm{inl}\, e) = j(e)$ and $F(\mathrm{inr}\, \star_\eta) = a$ (and analogously on incidences and claims). Injectivity splits into three cases: old–old equality cancels via $\mathcal{A}$'s constructive retraction; old–new equality contradicts preimage-freeness of a; and new–new has a single point $\star_\eta$. Uniqueness holds definitionally (`rfl`) because any morphism respecting old embeddings and the new generator is fixed on both constructors. □

Theorem 6.2 fixes the minimal geometry of conceptual expansion: $\mathcal{C}_\eta$ does not dictate a single future dynamics, but establishes **the unavoidable initial gateway through which every lawful expansion must factor injectively**. This addresses a regime distinct from classical

AGM theory change [20]: within a **fixed propositional language**, AGM partial-meet contraction $K \dot{-} p = \bigcap \gamma(K \perp p) \subseteq K$ uses a selection function γ over maximal non-implying subsets to relinquish implication of p, and when revising by a conflicting input p applies the Levi identity $K * p = \mathrm{Cn}((K \dot{-} \neg p) \cup \{p\})$ to contract $\neg p$ before adjoining p so as to restore consistency; Theorem 6.2 instead confronts **double emptiness of expression and realization in the existing language**, operating in zero axioms without choice or selection functions to expand the coordinate carrier while preserving all old expression theorems and prior open obligations intact.

6.2 Structurally Anchored Revisions and Shared First-Write Debt Succession

Definition 6.3 (Typed semantic revision and structural anchoring). For a rooted bankruptcy η, a **typed semantic revision** Ξ supplies: a non-conservative network translation (preserving old identities and open ledgers via constructive retracts while permitting preimage-free new coordinates); a new vocabulary and living root with terminal-handoff law; a new initial support retracting onto the old successor support; a claim-preserving, budget-nonincreasing embedding of the old target open ledger; a bidirectional emission-preserving occurrence correspondence between old successor and new initial visits; a **nonterminal canonical first write** (`CanonicalFirstRootWriteAt`) at the new root's initial visit; and a preimage-free zero-budget new direction row carrying semantic change such that **every inherited old row together with this new row forms the generated initial rows of that first write**.

Call Ξ **structurally anchored** (formally `FieldGrounded`) if it satisfies either: (i) **exact minimal anchoring**, recognizing the new network, law surface, translation, initial support, old-ledger embedding, new row, and semantic change field-for-field as $\mathcal{C}_\eta$ of Theorem 6.2; or (ii) **split-faithful anchoring**, where the new law surface is the new network's root semantics, the first-write compiler **actually selects** a strict split of one migrated old row into two distinct child rows of strictly smaller budgets, and no dormant rows exist outside the migrated ledger and new direction row.

Theorem 6.4 (Unique initial-direction factorization into the first write [6]). Every typed semantic revision Ξ induces an admissible coface revision whose unique faithful injection F_Ξ maps the minimal generator $\star_\eta$ onto the new direction row of Ξ consumed directly by Ξ's nonterminal first write.

Theorem 6.5 (Shared first-write authority succession and separate dispositions of old and new rows). Let Ξ be a structurally anchored typed revision for η. Then:

1. **Independent initial authorities**: the migrated old failure row $\mathrm{inl}(e)$ and the new direction row $\star_\eta$ each acquire initial causal entry authority from **their own** membership in the new initial generated patch;
2. **Passage through one shared first write**: both rows traverse the **same** compiler-generated nonterminal first-write successor equation to acquire target causal authorities;
3. **Separate dispositions sharing one canonical next current**: at the first-write target visit, the revised root's own causal "answer-and-next" compiler produces separate answers and continuations for the old failure row and the new direction row (and indeed for every generated initial row of the first write), all sharing **the exact same canonical next current**;

4. **No post-hoc failure-row substitution**: the migrated old row equals definitionally the demand row e at the old occurrence and the demand row of the source obstruction calculus's canonical event; the recovery interface exposes no parameter slot for submitting or swapping a failure row after the fact.

When Ξ is emitted at the exact occurrence by a fixed inquiry programme, this composite token is the inquiry's authoritative revision disposition.

Proof. Each initial generated row constructs its private-constructor initial causal authority from its membership in the initial patch; applying the unique nonterminal first-write continuation (`CanonicalFirstRootWriteAt` is a `Subsingleton`) once to each row yields both target authorities, on which the revised root's answer-and-next compiler evaluates. Because every row reads its next current from the same revised root at the same target visit, all next currents coincide definitionally. Item (4) composes Definition 6.1's demand-row and canonical-emission identities (Appendix C.5). □

Theorem 6.5 severs **registering an anomaly (dimensional expansion)** from **settling prior debt (actual fulfilment)** (Figure 4): writing the new direction alongside the entire old open ledger onto page one of the revised root grants expressive standing, never debt forgiveness. Whether the old obstruction is subsequently settled in the richer language and how the new semantic coordinate evolves remain two concurrent accounts, neither cancelling the other.

Example 6.6 (Two-step debt-preserving revision on a binary support [6]). Unfolding Example 1.2(c): on old support $\{\text{false}, \text{true}\}$ with open obligations of budget 2 at each point and double emptiness at true, step 1 emits an exact minimally anchored revision Ξ adjoining $\star_\eta$ with ledger "migrated old row (2), new row (0)" over $\mathbb{N}$-currents ($n \mapsto n+1$). The shared first write $0 \to 1$ delivers both rows ($2 \to 2$ and $0 \to 0$) to target visit 1, where the revised root's compiler disposes of each row and advances via $n \mapsto n+1$ to the shared next current 2; step 2 answers directly at visit 2 and advances to 3. Expressive capacity expands while the budget-2 historical obligation crosses the revision seam untouched.

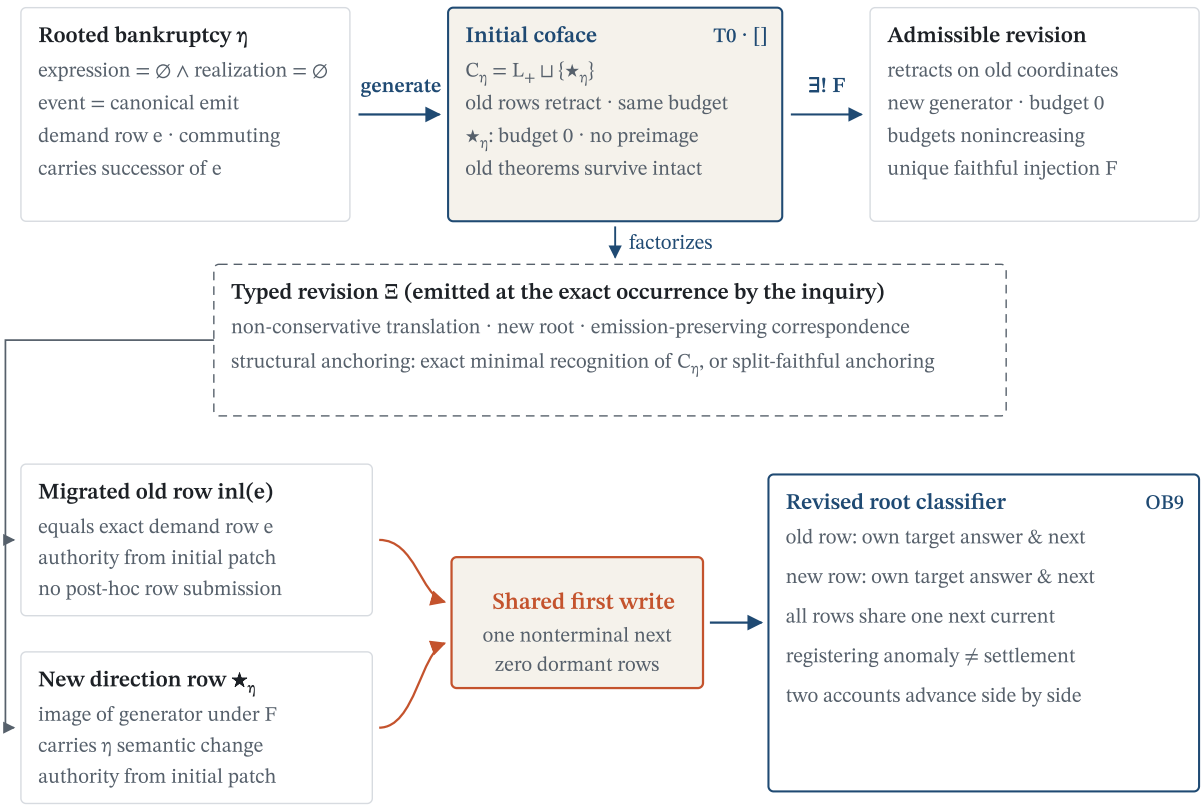


Figure 4 | Zero-axiom initial-coface extension and shared first-write succession. Top row: a rooted double bankruptcy η (empty expression and realization fibres, canonical emission aligned with demand row e) generates in zero axioms the categorical initial coface $C_\eta = L_+ \sqcup \{\star_\eta\}$, through which every admissible revision factors via a unique faithful injection F. Middle dashed box: a structurally anchored typed revision Ξ emitted at the exact occurrence by a fixed inquiry. Bottom row: the migrated old failure row $\mathrm{inl}(e)$ and the new direction row $\star_\eta$ obtain independent initial authorities, pass through the **same nonterminal first write** (vermilion), and receive separate dispositions sharing one canonical next current at the revised root; no arrow exists by which renaming absorbs or erases prior debt.

7 Long Analytic Windows and Full-Ledger Relay Under Inverse-Residual Correction

Same-debt payment and trace continuation scale intact from elementary polynomials to multi-stage operator-analytic computations (**OB10**). Along the finite-window analytic chain shared by Whole-Ledger Accounting [8] (Theorems 14.1–14.2) and the Riesz source calculus, the first window execution installs **four complete execution traces—forward stock, inverse residual, second mean, and cochain**—prior to decoding; the next correction frame reads the inverse residual η directly from that prior write, migrates all four historical traces intact, sets $s = \iota(\nu)$ and $r = \iota(\eta)$ (for initial band-limited source element ν and four-component embedding ι in (18)), substitutes the literal two-node correction $\kappa = \mathrm{const}\, s + (-\mathrm{id})(\mathrm{const}\, r)$ (syntax cost 2) into the complete window programme to generate fresh stock and correction traces alongside same-debt payment and canonical succession, and preserves request–boundary equivalence across all allowed future action words and sites.

This theorem holds **universally over its stated conditional input type**, which includes a zero-observation receipt z carrying the source zeta realization zero $\zeta_{\mathrm{src}}(z) = 0$, nontriviality $\forall n \in \mathbb{N},\ z \neq -2(n+1)$, and $\mathrm{Re}\, z > 1/2$ (if the Riemann Hypothesis holds, that input type is uninhabited and the theorem holds vacuously; we state it strictly as a universal wiring contract over that type without asserting inhabitation). Appendix E (Theorem E.1) supplies the complete analytic carriers, syntax budget $4k + 3$, and nested two-occurrence elimination.

8 Three-Layer Mechanically Checkable Failure Attribution and Intellectual Coordinates

A foundational principle earns its edge not by rendering itself unfalsifiable, but by pinning every claim to an explicit equality or type signature where a counterexample can strike (**OB11**). Here we specify the exact refutation conditions across three independent layers and situate our results relative to four classic traditions.

8.1 The Three-Layer Failure-Attribution Law

Any theoretical challenge or empirical discrepancy localizes to exactly one of three gates:

1. **Layer I: Structural identity, root-read identity, and universal formation recovery.** Targets: $\mathsf{RootRead}(o) = o$ and injective occurrence admission in Lemma 2.2, injective living currents in Theorem 2.8, left-inverse recoveries (7)–(8) in Theorem 4.2, and the finite-$\hbar$ pairing (9) in Theorem 4.4. Because all underlying declarations carry placeholder-free Lean 4 kernel proofs (`[]` at T0, standard three axioms at T1), a genuine hit at Layer I takes one of two forms: either exposing a translation discrepancy between our natural-language prose and the formal declarations in Appendix D (which corrects that prose statement and its downstream uses while leaving the formal theorem intact), or exhibiting a domain object that violates complete event-inventory admission in Definition 2.1 (noting that Theorem 4.2 already quantifies universally over **all** authoritative roots $\forall R$ on the `Type 0` interface).
2. **Layer II: Same-debt payment, paid-AST homological continuation, and shared first-write revision.** Targets: all-future equivalence (12), effect identity (14) with strict debit, double-negation paid-AST homomorphic recursion (15) in Theorem 5.4, the fourfold boundary–cochain identity (16) in Theorem 5.5, and the shared first write in Theorem 6.5. What is tested here is **strict object sameness**: does debit hit the original debt lineage rather than a newly opened row; do the homomorphic step and cochain differential consume this packet's actual paid AST and stepwise replay trace rather than a bare endpoint value; and do old and new rows traverse the very same first write? Constructing an admitted input on which a verification consumer reads a side-channel object refutes the corresponding Layer II claim.
3. **Layer III: Domain interpretation maps and empirical contracts.** Maps identifying formal source objects with physical spacetime fields, chiral fermions, molecular spectra, clinical neural signals, or social institutions—together with their blind empirical tests—are borne by the domain contracts of companion papers. An experimental discrepancy refutes and revises

that domain map or prediction; it is neither shielded by §2's zero-axiom no-magic theorem nor propagates backward to invalidate §5's syntax-payment theorems.

8.2 No Renaming a Refuted Equality to Escape into Revision

A strict methodological corollary accompanies these three layers: **a refuted equation may never be relabelled an "expressibility failure" so as to slip into the §6 revision channel.** Formally, entering Theorems 6.2 and 6.5 requires every field of Definition 6.1—doubly empty fibres, a canonical obstruction emission from the source calculus, an aligned open demand row, and commuting whole-ledger disposition. A false numerical identity or failed empirical prediction possesses none of those fields and must be corrected directly at the layer hit; conversely, by Theorem 3.3, an unverified coarse observation can never default to certified safety.

8.3 Exact Seams with Classical Origins and Modern Technical Frameworks

To situate each pillar precisely, we divide our primary-source comparison into two tiers (with consulted sections, axioms, and theorems documented in `literature.md`): Table 2 compares our approach against the four classical sources of sufficient reason, binary physical interrogation, mathematical communication, and proof-carrying admission; Table 3 compares our four main theorems pillar-by-pillar against modern frameworks in **dependent/univalent type theory and resource logics, quantum informational reconstructions and constructor theory, provenance semirings and finite-difference incrementalization, and AGM theory contraction and revision**.

Table 2 Comparison with four classical problem origins.

Primary work & thesis	Scope of original work	Shared problem seam	Constructive advance and exact distinction here
Leibniz, *Monadology* §§31–38 [1]: Sufficient Reason	Why any contingent fact holds thus and not otherwise; ultimate ground outside the series	Refusal to grant bare, groundless differences ontological standing	Posits no theological substance outside the world; realizes structural ground inside a zero-axiom kernel via injective occurrences $\mathsf{occ}(w)$ and living currents $\iota(s)$ (simultaneous relations or global constraints suffice without requiring earlier temporal causes)
Wheeler (1989) [2]: "It from Bit"	Physical "its" derive meaning from apparatus-elicited yes/no binary answers	Deep link between binary interrogation and physical standing	Our binary question tests *whether unaccounted differences can enter*, not single measurement outcomes; observers and apparatus live inside the same mother source $S_\star$, and unseen differences remain occurred (Theorem 3.3)

Primary work & thesis	Scope of original work	Shared problem seam	Constructive advance and exact distinction here
Shannon (1948) [3]: Mathematical Communication	Logarithmic entropy of random variables and channel coding capacity	The binary digit (bit) as minimal two-valued distinction	Strictly separates **binary standing**, **node-wise syntax cost**, and **statistical entropy**: a constant "no" has zero Shannon entropy, while zero numerical effect ($\epsilon = 0$) still pays positive syntax reduction steps (§5.3)
Necula & Lee (1996) [4]: Proof-Carrying Code (PCC)	Receiver reconstructs safety predicate from code and checks attached proof before execution	Execution standing paid on the spot by mechanically checkable proof rather than external trust	Upgrades one-shot code safety to a full-lifecycle source contract binding root occurrence, complete event inventory, strict same-debt debit, paid-AST cochain continuation, and shared first-write revision

Table 3 Pillar-by-pillar comparison with modern formal, physical, computational, and revision frameworks.

Pillar & modern framework	Primary works & core mechanism	Problem domain & premise structure in original works	Constructive advance and exact distinction here
I · Type identity & resource logics	**Martin-Löf** (1984) [10] Σ-projections; **HoTT** (2013) [11] Univalence; **Girard** (1987) [12] Linear Logic; **Atkey** (2011) [13] Amortised Separation Logic; system tool **Lean 4** [9]	[10] p. 32 adopts equality reflection; Univalence relates type equivalence and type equality $(A =_{\mathcal{U}} B) \simeq (A \simeq B)$ within a universe; [12] controls copying/discarding of marked formulas via exponentials; [13] verifies heap resources via LP-inferable credits and fallible environmental `acquire` (§3.6)	Inside an intensional kernel with T0 empty axiom closure `[]` certified by our audit receipts, state identity is determined by single-constructor unfolding and registered occurrences $\mathtt{occ}(w)$; syntax credit is fixed endogenously by the residual AST with strict non-refill debit and an 8-vector `#guard_msgs` firewall
II · Physical reconstruction & refinement	**Chiribella et al.** (2011) [14] 5 axioms + Purification; **Deutsch & Marletto** (2015) [15] Constructor Theory; **Abadi & Lamport** (1991) [16] Refinement Mappings	[14] adopts an operational-probabilistic framework plus purification to derive quantum theory; [15] states laws via possible/impossible tasks and Interoperability (Principle III); [16] augments lower-level specifications under regularity conditions with behaviour-preserving history and prophecy variables	T0 posits no primitive impossibility modality yet derives `noMagicBit`; T1 requires neither operational-probabilistic primitives, interoperability postulates, nor specification augmentation, synthesizing separating addresses internally on `Type 0` from one deterministic `SpinPair` mother $S_\star$ with left-inverse recovery (Theorem 4.2)

Pillar & modern framework	Primary works & core mechanism	Problem domain & premise structure in original works	Constructive advance and exact distinction here
III · Provenance & finite differences	**Green et al.** (2007) [17] Provenance Semirings $\mathbb{N}[X]$; **Cai et al.** (2014) [18] Static `Derive` Thm. 3.11; **Budiu et al.** (2023) [19] DBSP §4.1 $\mathbb{Z}$-sets & Thm. 3.4	[17] factorizes positive relational query evaluation via polynomial semiring homomorphisms; [18] and [19] give static differentiation on general change structures and incremental calculus on streams of abelian-group values, respectively. Abelian groups give a special case of the former	Retains the exact finite-difference expansion (10), while turning literal residuals into self-paying endogenous requests ($\epsilon = 0$ still pays 2 steps), recovering paid ASTs via double negation for continuing method f (OB12), and replaying all 9 step constructors to generate stock boundaries and $d^{0,1}$ (OB13)
IV · Theory contraction & belief revision	**Alchourrón, Gärdenfors & Makinson** (1985) [20]: partial-meet contraction $K \dot{-} p = \bigcap \gamma(K \perp p)$ and Levi identity	Within a **fixed propositional language**, partial-meet contraction uses selection function γ to drop implication of p ($K \dot{-} p \subseteq K$); revising by a conflicting input p applies the Levi identity to contract $\neg p$ before adjoining p to restore consistency	Confronts **double emptiness of expression and realization** in **zero axioms (no selection function)**, generating the categorical initial coface $L_+ \sqcup \{\star_\eta\}$ (Theorem 6.2) while carrying prior open obligations intact through one shared nonterminal first write without contraction (Theorem 6.5)

9 Conclusion: Four Payouts of Ontological Rigidity and Companion Interfaces

We return to our opening question: can a real difference enter reality without accounting for any structural provenance? Without circularity or external prohibitory axioms, four constructive results establish the negative answer and put it to work.

In the zero-axiom constructive kernel (T0), single-constructor unfolding and lawful admission establish $\mathsf{RootRead}(o) = o$, endowing every real difference across any authoritative root or terminal/revision-crossing living process with positive structural identity, constructively emptying external supplements and magic bits while sealing eight bypass surfaces at compile time. At the analytic completion layer (T1), pointwise uniform completion of finite programmes from the single primitive-event-generated spin-pair mother $S_\star$ universally forms all lawful worlds, complete inquiries, and macro-processes on the `Type 0` interface—eliminating any independent second base source—and directly realizes the four-dimensional $\mathrm{Spin}^+(1,3) \times \mathrm{SU}(7)$ nine-field configuration and exact $4\sqrt{2}$ quantum pairing. Along dynamic continuation and language revision, all-future residuals endogenously birth requests and strictly debit the same syntax debt, double-negation extraction recovers paid ASTs to drive continuing homomorphic methods across cofinal stops, primitive-by-primitive word replay updates stock boundaries and the cochain differential $d^{0,1}$ in lockstep, and rooted double bankruptcy generates in zero axioms the categorical initial coface $L_+ \sqcup \{\star_\eta\}$, compelling every anchored revision to carry prior obligations intact through one shared nonterminal first write.

As the ontological and methodological gateway to the wider research programme, this paper hands specialized obligations cleanly to its companions: four-tier process algebra, exact free-module evaluation kernels, and bidirectional step–request interval inversion to *Source-Native Process Algebra* [6]; three-law responsibility conservation, capacity-shortage positive obstruction, and constant-ledger relation inversion to *Whole-Ledger Accounting* [8]; and ten-stage continuous–quantum field closure with zero-knob low-energy deduction to the third edition of the physics flagship [7] and its phenomenological series. Every remaining open obligation (domain-specific interpretation maps, blind experimental tests, and tomorrows not yet paid for by an actual run) stays explicitly named on the books, neither prematurely declared settled nor allowed to slip off the ledger.

Appendix A Zero-Axiom Total Reality, Root-Read Identity, and Observer Safety Proofs

This appendix provides self-contained proof details for §§2–3 in the notation of the main text. Aside from the complexified torsion instance of Theorem 3.3(3), every declaration resides in the zero-axiom kernel (`#print axioms = []`) of the fixed verification source package (source Y in Appendix D).

A.1 Single-Constructor Occurrence Unfolding and the Root-Read Mother Law

In `LivingLawRootedAccountedUnfoldingKernel.lean`, the mutually inductive types `RootedAccountedUnfolding` and `AccountedBranches` admit the sole occurrence constructor `.occur root branches` (with `.nil` and `.cons` on branch lists). Structural recursion establishes `map_id` ($o.\mathrm{map}(\mathrm{id}) = o$) and `map_map` ($(o.\mathrm{map}\, f).\mathrm{map}\, g = o.\mathrm{map}(g \circ f)$). Because `frontier` returns `[r]` on leaves `.occur r .nil` and concatenates child frontiers otherwise, mutual induction with `append_ne_nil_of_left_ne_nil` proves in zero axioms `frontier_ne_nil` ($o.\mathrm{frontier} \neq [\,]$). One-step unfolding `advance next` attaches `.singleton (next r)` at each open leaf, yielding `frontier_advance` and `advance_assoc` via `flatMap_append` and local list associativity. Universal evaluation `fold atOccurrence` recurses over `.occur`; any candidate evaluator commuting with `.occur` satisfies `fold_unique : cand o = o.foldatOccurrence` by mutual induction.

In `LivingLawRootCoreRepresentationKernel.lean`, `LivingLawRootTemporalAnswerNextKernel.lean`, and `LivingLawRootInquiryFoundationKernel.lean`, on complete temporal causal root occurrences `rootCofaceRead (o : ExactTemporalCausalRootEvent root) := o` is the literal identity map, proving in zero axioms universal consumer safety `rootCofaceRead_consumerSafe` and emptiness of hidden differences `no_hiddenNull_at_rootCofaceRead` for every downstream consumer; for fixed `(root, visit)`, `SourceNativeTemporalVisitGeneratedEvolutionAt root visit` has the sole constructor `.canonical`, so `occurrence_eq`, `priorPatches_eq`, `currentPatch_eq`, and `wholeLedgerWriteBack_eq` all hold by `rfl` / `HEq.rfl`. On `SourceNativeRootSemanticFaceAt`, `rootRead` evaluates `project projection (root.emitted visit.current) active` directly, while `SourceNativeInquiryAnswerConsumerAt.installedAuthority_factorizes` binds the consumer receipt heterogeneously via `project_heq` to `(query, event, entry, answerFace.rootRead)`, closing Lemma 2.2(1)'s $\mathsf{RootRead}(o) = o$ at the machine level.

A.2 Temporal Reachability Induction and the Compile-Time Firewall

Given R, `SourceNativePostCofinalReachableAt` has constructors `.cofinal(u)` and `.step(h, eq)`, while `SourceNativeTemporalReachableAt` has constructors `.finite(h)` and `.postCofinal(h)`. Because post-cofinal histories start solely from `.cofinal`, finite visits never cross into the post-cofinal branch; destructuring `.finite(v) = .finite(v')` field by field proves `finite_injective`. In Proposition 2.3(1), v_1 (initial) and v_2 (one-step) have disjoint constructors, forcing inequality.

Compile-time guards in Proposition 2.3(3)–(4) operate at two levels: `LivingLawZeroLawRootAdmissionRegression.lean` (`HOmework.Checks.OneBit.ZeroLawRootAdmissionRegression`) verifies via `fail_if_success` that passing a `Bool` argument to `finite` or coercing an unadmitted

`mixedTerminalRoot` into a lawful state fails type-checking, with six `#print axioms` checks confirming `[]`; `Tracks/LivingLawSingleRootAuthorityFacadeRegression.lean` imports all twenty semantic regression suites and asserts via eight `#guard_msgs` checks that `U8RevisionCalculus`, `GeneratedRevisionFamilyAt`, `RevisionSelectionCalculus`, `U7ObstructionLifecycle.demandEvolutionRun`, `TheoryRevisionLifecycleRegression.settled`, `SourceNativeGeneratedU8RevisionAuthority`, `SourceNativeGeneratedU8SettlementAuthority`, and `SourceNativeOperationalSearchRecognitionAt.exhaustive` remain absent from the public authority namespace.

A.3 Root-Level and Cross-Handoff Living-Process Total Reality

In Theorem 2.6's semantics (5), actuality takes `generatedAtTemporalVisit_isCanonical` on both visits, stable reference takes the dependent pair $\langle \mathsf{occ}(w), \mathsf{rfl} \rangle$, and `RootTotalReality.isTotal` returns `Or.inr (registeredOccurrence_ne_of_ne actual.realDifference)`, chaining into Theorem 2.5's `noExternalSupplement` and `noMagicBit`. In Theorem 2.8, `RootProcessTotalReality` compares complete three-field living currents `process.stateAt s` (vocabulary, handoff-bearing living root closure, and temporal visit), derives `registeredCurrent_ne_of_ne` from `stateAt_injective`, and excludes magic bits across local terminals and theory revisions in zero axioms, accepting no external `NoMagic` or classifier parameter.

A.4 Current Reflection and Law-Surface Preservation

In Definition 2.7, the generated-successor relation is a single conjunctive proposition coupling erased target equality with heterogeneous living-root preservation on non-terminal branches, preventing detached reuse on terminal branches. In Theorem 3.1(1), injectivity reduces $\iota(s) = \iota(s')$ to $s = s'$, after which both `canonicalCausalAnswerAndNext` terms coincide (inside a single-constructor carrier); mapping $\rho \mapsto \rho.\mathrm{lawSurface}$ across the successor equation and applying one-step root law-surface preservation proves Theorem 3.1(3).

A.5 Observer-Safety Refutation and Global Integral Torsion Annihilation

In Theorem 3.3(2), scoped visibility filtering drops an invisible-scope commitment while complete slot observation grows by one slot, directly populating `FullTraceRedirect`. In Theorem 3.3(3), $G = \{(g_n) \in \prod_n C_n : \rho_n(g_{n+1}) = g_n\}$, $F(g) = (1 \otimes_{\mathbb{Z}} \pi_n(g))_n$, and $J(g) = \pi_1(g)$. Whole-Ledger Accounting [8] (Theorems 13.1–13.3) constructs $z \in G$ with $J(z) = z_1 \neq 0$ from the unit arithmetic root, while finite-order torsion satisfies $1 \otimes_{\mathbb{Z}} z_n = 0$ stagewise over $\mathbb{C}$ without commuting limits past tensors, giving $F(z) = 0 = F(0)$ and refuting $J = h \circ F$ for arbitrary h.

Appendix B Proof of Universal Three-Domain Formation from $S_\star$

B.1 From Finite Observation Approximation to Bijective Uniform Completion

Let $\mathcal{A}_r$ separate points of X_r. For any finite $F \subset X_r$, choosing one separating address per distinct pair yields a finite tuple $(a_0, \dots, a_{n-1})$ making $x \mapsto (\mathrm{obs}_r(a_i, x))_{i<n}$—and its zero-padded stream `pad_n`—injective on F. Prescribing values of any target $f : X_r \to \mathbb{S}$ on that finite image and invoking the finite-evaluator approximation theorem of $S_\star$ [7] (Appendix C.9) yields, for every $b \in \mathbb{N}$ and $\varepsilon > 0$, a mother visit at index $10 + c$ (c the programme code) and $p \in \mathsf{Prog}_r$ satisfying

$$\left|L_r(p, x)(j) - f(x)(j)\right| < \varepsilon \qquad (\forall x \in F,\ 0 \le j < b). \tag{B.1}$$

Because (F, b, ε) indexes a fundamental system of entourages for the pointwise product uniform structure on $X_r \to \mathbb{S}$, $\mathcal{R}_r$ is dense. The completed extension $\mathsf{read}_r : \mathcal{M}_r \to (X_r \to \mathbb{S})$ is a uniform embedding whose image is complete—hence closed in the complete Hausdorff space $X_r \to \mathbb{S}$—and contains the dense subset $\mathcal{R}_r$, proving read_r bijective and ensuring that every $m \in \mathcal{M}_r$ satisfies Definition 4.1.

B.2 Unified Disjoint-Sum Witness and Left-Inverse Parent Restriction

For any macro-process P on `Type 0`, form the `Type 0` disjoint union

$$\mathcal{U}_P = \mathrm{State}(P) \ \sqcup \coprod_{s \in \mathrm{State}(P)} \mathrm{Query}(P.\mathsf{stateAt}(s)). \tag{B.2}$$

Cumulative material construction extracts a sufficient rank and faithful point-separating addresses internally from $\mathcal{U}_P$ (keeping distinct coordinates for answered, unvisited, and observationally tied states), forming per-state query nodes alongside state-domain and successor graphs. Inquiry formation similarly encodes all five branches of query compilation (answer, action, and revision clauses plus routing), while world formation pairs root and visit materials. Embedding component materials into a common rank via bijective stream interleaving guarantees that stream unpairing provides exact left-inverse parent restrictions across the entire joint material space.

B.3 Canonical Realization and All-Depth Trajectory Transport

Reading state carrier, complete node family, event addresses, and successor graph from the joint material m via B.2's left-inverse identities produces a state-relabelled process P_m whose restriction along the generated state equivalence satisfies $\mathsf{restrict}(P_m) = P$ in (7). Record equality (7) then transports per-state query compilations, engines, activations, complete activation laws, and discrete steps across every depth $n \in \mathbb{N}$ to yield (8), while Theorem 2.6 endows each formed world's recovered root with total reality.

B.4 Non-Bypass Audit on Five Downstream Verification Chains

In `MotherFixedSourceRealizationConsumer.lean` (Corollary 4.3), the five declarations `original_world`, `original_inquiry`, `original_process`, `original_physical_seal_and_whole_law`, and `o`

`riginal_physical_tick_and_history` each invoke `root_admission_fixed_mother_complete_realization` directly, recovering universal worlds, inquiries, and processes as well as physical state 16, step $16 \to 17$, arbitrary-depth n histories, mother visit 10, and `SameOccurrenceActivation`. Automated proof-term audits confirm direct invocation of the master declaration, zero direct references to the three low-level coverage or four low-level recovery lemmas, zero unsafe/partial constants, and an axiom closure strictly equal to `{propext, Classical.choice, Quot.sound}`.

Appendix C Proofs of Same-Debt Payment, Paid-AST Continuation, and Initial-Coface Revision

C.1 Invariance of the All-Environment Zero Submodule Z (OB8)

In the free $\mathbb{Z}$-module $\mathrm{Fm}(V, X; s)$, $Z = \{w : \forall \Theta,\ \mathrm{ev}_\Theta(w) = 0\}$ is a submodule. By (10), $\mathrm{ev}_{\Gamma+\delta} = \mathrm{ev}_\Gamma + \mathrm{eff}_{\Gamma,\delta}$, so $w \in Z$ implies $(\mathrm{ev}_\Gamma(w), \mathrm{eff}_{\Gamma,\delta}(w)) = (0,0)$ for all (Γ, δ). Structural induction shows that `liftExpr`—sending constants $v \mapsto (v, 0)$, unary $g \mapsto (g, g)$, and bilinear $B \mapsto (B(u,v), B(u,dv) + B(du,v) + B(du,dv))$—evaluates under $\mathrm{pairEnv}(\Gamma, \delta)$ to $(\mathrm{ev}_\Gamma, D(\Gamma, \delta))$. Because every pair environment equals the `pairEnv` of its projections, `liftExpr`, language transport, and packet transport all preserve Z. Abelian cancellation places $[r_m] - ([a] - [p]) \in Z$, and word induction yields (12).

C.2 Positive Syntax Clock and Unique Same-Debt Lineage (OB8)

Stepwise run induction gives $c(\mathrm{state}(b)) = c(b.\mathrm{AST}) - (b.\mathrm{depth} + 1) = Q - 1$. Since $Q = c(a) + c(p) + 2 \geq 2$ by (11), $Q - 1 > 0$, ruling out zero-cost mathematical settlement and forcing a paid reduction step with strict unit debit. Sum-type disjointness between original mathematical debt (left injection) and inherited rows (right injection) locks a unique same-debt target, and induction over finite visits identifies the payment transition heterogeneously with the whole-ledger destination (Theorems 5.2–5.3).

C.3 Double-Negation Paid-AST Recovery, Homomorphic Step, and Callback Stock (OB12)

We prove Theorem 5.4 (source MV5 in Appendix D, `PaidMethod/{Source,Recovery,Admissions}.lean`):

1. **Prefix input invariance and double-negation extraction (`after_native_retained`)**: for $\mathsf{pkt} = (n, (\mathrm{source}, \mathrm{seed}), \mathsf{cfg}, \langle \mathsf{rfl} \rangle)$, the receiver frame `native` registers $e_0 = N.\mathrm{expression}(\mathrm{actualMaterial}(\mathrm{source}, \mathsf{cfg}))$. Let $d = (\mathrm{receipt}(\mathsf{stockCfg}, \mathrm{native}, 0)).1$ be the local completion distance. For $0 \leq m < d$, `receipt_paid_prefix` forces the `.paid` branch `mathNext`, keeping `registered.input.expression` invariant (`paid_prefix_expression`) up to `selected` at $m = d$. At settlement, `nextBorn` registers `Expr.add e0 (Expr.linear (-id) selected.event.state.1)` (writing `e0` for e_0), on which pattern-matching `retainedCode` extracts the right summand and applies `-id`, giving

$$\mathsf{retainedCode}(\mathrm{next}) \doteq (-\mathrm{id})((-\mathrm{id})(\mathrm{actualMaterial}(\mathrm{source}, \mathsf{cfg}).\mathrm{state}.1)).$$

2. **Value recovery and homomorphic recursion (`after_native_value`, `after_native_whole_action`)**: owner-free `paid_expression` identifies that paid-state AST with `.const v_k`; abelian involution `-(-x)=x` (`negate_paid`) evaluates `retainedCode` to v_k in any environment Γ'. At level $n+1$, `raw f (n+1)` evaluates `.linear (pairAction (actionAt f (n+1))) (liftExpr (retainedCode ...))` under $\mathrm{pairEnv}(\Gamma', 0)$. Since `effect_zero` gives `expr.effect env 0 = 0`, `pair_zero` yields `(v_k, 0)`, and applying `pairAction` with `map_zero` yields `(actionAt f (n+1) v_k, 0)`, proving (15).
3. **Callback stock containment and unified face `RootRead` (`actual_complete_query_written`, `cofinal_paid_method_faces`)**: setting `nextPairInventory := some (Stock.preserve prior completeQueryWritten)` routes every trace event into `birthStock = Stock.preserve frame.pairInventory (Stock.preserve callback generatedPairStock)` via `parallel_left` and `parallel_right`. Sevenfold coface composition installs `actualFace` before emission with `packet := actualFace.rootRead` (`source_readback := HEq.rfl`), generating `UnifiedFourFace` (algebraic, combinatorial, topological, and logical faces together with relation and cochain faces) and dual readbacks; on the Hilbert-space and obstacle-calculus runtime of `Admissions.lean`, strictly monotone cofinal indices `D.index` and `field_segment` close arbitrary finite horizons `finite_paid_method_faces`.

C.4 Primitive Step Replay and the Fourfold Boundary–Cochain–Coimage Identity (OB13)

We prove Theorem 5.5 (source MV5 in Appendix D, `WordExecution.lean` and `RelationCochain/{Source,Consumer}.lean`):

1. **Contextual step embedding and exact charge (`WordExecution`)**: for any complete living source process carrier $w \in \mathrm{Carrier}(\mathcal{P})$ and `readEnv`, structural induction verifies `embed_eval` and `binding_word`. Recursing over all nine constructors of `Step (observerEnv readEnv w) before after` defines `embeddedStep` (routing `.bind` through contextual evaluation and `embed_eval`, mapping constant reductions to `Trace.single`, and lifting congruences recursively), concatenating to `embeddedTrace` and `pairedForward`. `length_to_const` immediately yields `forward_charge : (forward readEnv w expr).length = remaining ((embed readEnv expr).subst binding)`.
2. **Containment in born stock and next query (`native_replay_written`, `native_replay_next_query`)**: at runtime offset $11 + \mathrm{count}$, inductive invariant `actual_grounded` establishes `environment_actual` and `paid_current`, transporting `nativeReplayAt` along proven environment equalities (never conflating traces via endpoint values) so `exposure(nativeReplayAt) = replayStock`. Applying `birth_keeps_callback` and `complete_written_preserves` places both the original paid trace and replay exposure inside `birthStock.trace` and the next born frame's `completeQueryWritten.trace`.
3. **Fourfold differential identity, unfolding erasure, and same-stop physical readback (Equation (16) and `actual_relation_cochain_method`)**: applying telescoping trace-boundary evaluation $\mathsf{relMap}_E(\tau.\mathrm{relationWords}) = [e_{\mathrm{in}}] - [e_{\mathrm{out}}]$ to `nativeReplayAt` under E_{stop} and `nativeSourceAt` under $E_{\mathrm{src}} = E_{\mathrm{stop}+1}$ together with `SubstitutionLift.expression` proves `actual_boundary_from_source` ($\partial_{\mathrm{act}}.\mathrm{val} = \mathsf{subst}_{\mathrm{pair}}(\mathsf{relMap}_{E_{\mathrm{src}}}(\mu_{\mathrm{stop}}))$). Chaining `SourceSubstitution.boundary_square` gives `actual_complex_boundary` ($\mathsf{relMap}_{E_{\mathrm{stop}}}(\tau_1(\mu_{\mathrm{stop}})) = \partial_{\mathrm{act}}.\mathrm{val}$), and `natComplex_d_zero_one` gives `generated_differential` ($(d^{0,1}_{\mathcal{C}}(\partial_{\mathrm{act}})).\mathrm{val} = \partial_{\mathrm{act}}.\mathrm{val}$), establishing

(16), while `completion_kills_generated_differential` proves vanishing under completion projection. `paired_binding_square` constructs the unified-face morphism `faceAction` yielding `actual_coimage_boundary` and `actual_coimage_differential`; finally, `map_map`, `map_id`, and `root_map` on `sourceOccurrence` prove `(sourceOccurrence.map Prod.fst) = sourceUnfolding` and `sourceOccurrence.root.2 = face.rootRead`, combined with `physicalReadback` (reading back next-tick `((sourceState, forcedTrace), 0)`) and same-stop `runFrom` segment splicing to close `actual_relation_cochain_method`.

C.5 Zero-Axiom Initial-Coface Universal Property and Shared First-Write Chain (OB9)

In `LivingLawObstructionGeneratedMinimalCofaceKernel.lean`, `ConstructiveRetract A B` requires only `backward (forward a) = a`, admitting preimage-free coordinates in B; `FaithfulCofaceMorphism` has the sole constructor `.canonical`, so `cases morphism; rfl` proves `minimal_coface_factorization_unique` in zero axioms. In `LivingLawTypedSemanticWorldNetworkU8RecoveryKernel.lean`, Theorem 6.5's authority pipeline maps

$$\text{initial-patch membership} \longmapsto \text{initial entry authority} \longmapsto \text{shared first-write target authority} \longmapsto \text{revised-root answer and next},$$

reusing the single `CanonicalFirstRootWriteAt` proof on both old and new rows and evaluating the revised root's compiler at one target visit so all next currents coincide definitionally; under split-faithful anchoring, distinct child rows strictly decrease `progressBudget` relative to their common migrated parent.

Appendix D Machine-Verification Index and Fixed Source Map (OB1–OB13)

The table below maps all stable claims **OB1**–**OB13** to their bilingual text locations, fixed research declarations, and public verification entry points.

Claim	Text & appendix	Fixed-source modules and core declarations	Public package & independent audit
OB1 Binary standing	§1.1, Def. 2.4	Y: `LivingLawTotalRealityKernel` (`MagicBitAt`); Shannon (1948)	Direct source & primary-text inspection

Claim	Text & appendix	Fixed-source modules and core declarations	Public package & independent audit
OB2 Ghost bit & firewall	Ex. 1.2(a), Lem. 2.2, Prop. 2.3, A.1–A.2	Y: `LivingLawRootedAccountedUnfoldingKernel` (`occur`, `frontier_ne_nil`, `fold_unique`); `LivingLawRootCoreRepresentationKernel` (`rootCofaceRead`); `LivingLawRootTemporalAnswerNextKernel` (`occurrence_eq`, `wholeLedgerWriteBack_eq`); `LivingLawRootInquiryFoundationKernel` (`rootRead`); `LivingLawZeroLawRootAdmission{Kernel,Regression}` (`ghostFalse`, `no_ghost_argument_in_world_state_producer`); `Tracks/LivingLawSingleRootAuthorityFacadeRegression` (8 `#guard_msgs`)	H0 `HOmework.Foundation.{Source.AccountedUnfolding,Cofinal.TemporalAnswer,Inquiry.Protocol}`; `HOmework.Checks.OneBit` (`make check-one-bit-full`, `[]`). Eight guards in the public firewall
OB3 Zero-axiom total reality	Main Thm. 1.1(1), Def. 2.4, Thms. 2.5–2.8, Ex. 2.9, A.3	Y: `LivingLawTotalRealityKernel` (`toExternalSupplement`, `noExternalSupplement`, `noMagicBit`); `LivingLawZeroLawRootAdmissionKernel` (`registeredOccurrence_ne_of_ne`); `LivingLawRootTotalRealityAdapter` (`RootTotalReality.isTotal`, `RootProcessTotalReality.isTotal`) & regressions	H0 Whole-Ledger `WA1` (`HOmework.Foundation.{Authority.ZeroLawAdmission,Semantics.RootReality}`, `[]`)
OB4 Rigidity & history	Thm. 3.1, A.4	Y: `LivingLawRootCoreRepresentationKernel` (`canonicalAnswerAndNext_heq_of_current_eq`); `LivingLawRootAnswerNextHistoryKernel` (`successor_difference_reflects_current`, `successor_lawSurface_eq`, `generateHistory`)	H0 Process Core `C3` & Whole-Ledger (`[]`)
OB5 Observer safety & torsion	Def. 3.2, Thm. 3.3, A.5	Y: `ResponsibilityObserverKernel` (`ConsumerSafe`, `HiddenNullWitness`, `FullTraceRedirect`); `NativeReceiptGlobalCompression`, `NativeReceiptIntegralConsumerSeparation`	H0 Whole-Ledger `WA6`, `WA11`

Claim	Text & appendix	Fixed-source modules and core declarations	Public package & independent audit
OB6 Universal $S_\star$ formation	Main Thm. 1.1(2), Def. 4.1, Thm. 4.2, B.1–B.3	F: `Formation`, `CompleteRealization` (`root_admission_fixed_mother_complete_realization`, `low_formation`, `high_formation`)	H0 First-Release `phys.P22` (Std3)
OB7 Unified-field check	Cor. 4.3, Thm. 4.4, B.4	F: `MotherFixedSourceRealizationConsumer` (5 non-bypass checks); H: `PhysicsCore/Stage10/Realization` (`source_realization_consumed`)	H0 First-Release `phys.P6`, `phys.P22` & Physics v3 `a5e774a4`
OB8 Residual & payment	Main Thm. 1.1(3), Ex. 1.2(b), Thms. 5.1–5.3, C.1–C.2	MV4: `RootConsumer/{Semantics,Source,Consumer}` (`actual_root_future`, `actual_root_eliminate`); `RootedEffect/{Source,Consumer}` (`consume_rooted`); `Completion/{Source,Consumer}` (`consume_rooted_completed_answer`)	H0 Whole-Ledger `WA38`, `WA39` (`Versions.Rba7e274b`)
OB9 Initial coface & revision	Main Thm. 1.1(4), Ex. 1.2(c), Defs. 6.1, 6.3, Thms. 6.2, 6.4–6.5, Ex. 6.6, C.5	Y: `LivingLawObstructionGeneratedMinimalCofaceKernel` (`root_obstruction_generates_minimal_coface`, `minimal_coface_initial`, `minimal_coface_factorization_unique`); `LivingLawTypedSemanticWorldNetworkU8{,Lifecycle,Recovery}Kernel` & regressions (all `[]`)	H0 Whole-Ledger `WA9` (`HOmework.Foundation.Inquiry.{SemanticRevision,RevisionRecovery}`, `[]`)
OB10 Long-window relay	§7, App. E Thm. E.1	MV4: `ClozelOriginalKFinitePaWindowOperatorSourceAction/{Consumer,Complete}`; `ClozelOriginalKWindowInverseResidualSourceAction/{Received,Consumer}`	H0 Whole-Ledger `WA40` (declaration-closure audit PASS)
OB11 Failure attribution	§§8.1–8.3, Tables 2–3	Input types, output equalities, and paired controls across OB2–OB10, OB12–OB13	Methodological synthesis & `check_release.py`

Claim	Text & appendix	Fixed-source modules and core declarations	Public package & independent audit
OB12 Paid-AST method f	Main Thm. 1.1(3), Thm. 5.4, Ex. 5.6, C.3	MV5: `PaidMethod/Source` (`retainedCode`, `factory`, `templateFourFaces`) → `Recovery` (`after_native_retained`, `after_native_whole_action`, `actual_complete_query_written`) → `Admissions` (`cofinal_paid_method_faces`, `finite_paid_method_faces`)	second-edition claim map, CP483
OB13 Step replay & $d^{0,1}$	Main Thm. 1.1(3), Thm. 5.5, Ex. 5.6, C.4	MV5: `Context/WordExecution` (`embeddedStep`, `embeddedTrace`, `forward_charge`) → `RelationCochain/Source` (`replayStock`, `replay_written`, `replay_next_query`) → `Consumer` (`actual_complex_boundary`, `actual_complex_differential`, `actual_coimage_differential`, `physicalReadback`, `actual_relation_cochain_method`)	second-edition claim map, CP485

Fixed research commits in Homework: $\mathbf{Y}$ = `b81a443863f4d7135ebaca92c5bb7d208ccc18ca`, $\mathbf{H}$ = `e60a86058abcb74b7f2225f2383c182be8343785`, $\mathbf{F}$ = `cde4b910929d9bc57554aa56f6ade116bc3b8675`, $\mathbf{MV4}$ = `ba7e274beb1de38900a6fc2c8c8dfe6cfad9d9aa`, $\mathbf{MV5}$ = `df421ff600085cbc290a7ae946193961bed7c83b`. Exact SHA-256 digests and declaration signatures are verified automatically by `release-selection-v2.json` and `scripts/check_release.py`.

Appendix E Complete Conditional Wiring Contract for Long Analytic Windows

This appendix states the multi-stage analytic window wiring contract summarized in §7 (**OB10**, Whole-Ledger Accounting [8] Theorems 14.1–14.2). Underlying functional-analytic operators—the complex Hilbert space, defect and Fourier operators, Pa qualification projection, centered Möbius cutoffs, and two mean identities—come from the Riesz source calculus; here we state the exact syntax cost, four-trace cross-frame migration, same-debt payment, and all-future equivalence.

E.1 Four-Component Analytic Carrier and Conditional Input Domain

Let $\mathcal{H}$ be the original complex Hilbert carrier, D the defect operator, and $\mathcal{F}$ the Fourier operator, with four-component product and left-invertible embedding

$$\mathcal{B} = (\mathcal{H} \times \mathcal{H}) \times (\mathcal{H} \times \mathcal{H}), \qquad \iota(f) = \big((f, Df), (\mathcal{F}f, \mathcal{F}Df)\big), \qquad \pi \circ \iota = \mathrm{id}_{\mathcal{H}}. \tag{18}$$

Set the band-limited source element $\nu = M\mathcal{E}\pi(S)$ for Paley–Wiener qualification projection $\mathcal{E}$ [8], qualification multiplier M, and input $S \in \mathcal{B}$; write Z_a for cutoff zero-extension, P for the first-pass inverse operator, and R_a for interval readout restriction. The finite cutoff window operator is

$$W_a(f) = \sum_{m \in L_a} \frac{1}{m} A_{a,m} f - \frac{1}{2} \ell(Jf)\, c_a, \tag{19}$$

with centered Möbius cutoff list L_a, scale pull-backs $A_{a,m}$, window integral ℓ, Tate reciprocal involution J, and constant interval field c_a. The conditional input type comprises a zero-observation receipt z on the analytic occurrence (carrying source zeta realization zero $\zeta_{\mathrm{src}}(z) = 0$, nontriviality $\forall n \in \mathbb{N},\ z \neq -2(n{+}1)$, and $\mathrm{Re}\, z > 1/2$) together with arbitrary depth, ordinal, and real parameters h, H, N, a. Retaining summation, reciprocal weights, and the Tate pole as explicit AST nodes fixes the window syntax budget at $4k + 3$ when $|L_a| = k$. Theorem E.1 holds universally over this conditional input type (and holds vacuously if the Riemann Hypothesis is true).

E.2 Cross-Frame Four-Trace Migration, Residual Fuel, and Joint Elimination

Theorem E.1 (Four prior traces enter inverse-residual correction, same-debt payment, and every future [8]). For every original input in the domain of E.1:

1. **First-frame execution and inverse-residual readout**: the first actual write difference satisfies $\Delta_F = \iota Z_a W_a(\nu)$; forward stock, inverse residual, second mean, and cochain inventories are installed before birth; reading directly from that write gives

$$\eta = \nu - MP\pi(\Delta_F), \qquad s = \iota(\nu), \qquad r = \iota(\eta), \tag{20}$$

   with η equal to the complete residual generated by the first inverse programme, and the birth completion receipt supplies its writer, strict same-debt payment, and exact whole-ledger destination.
2. **Literal correction and window-difference identity**: the next frame executes $\kappa = \mathrm{const}\, s + (-\mathrm{id})(\mathrm{const}\, r)$ (2 charged syntax nodes) inside the complete window programme to produce the correction packet, completion receipt, and whole-ledger write satisfying

$$\Delta_C = \iota Z_a W_a(\nu - \eta), \qquad R_a\pi(\Delta_C) - W_a(\nu) = -W_a(\eta). \tag{21}$$

3. **Nested six-trace elimination**: the four historical traces of frame one migrate intact into the correction packet alongside four newly generated window traces plus fresh stock and correction traces; prior write, new write, new whole-ledger payment, environment/AST identities, and

execution traces eliminate jointly in one verification consumer without conflating inventory maintenance with original debt budget.

4. **True source-seed lock and all-future invariance**: birth root and current match the complete packet's root and current, continuation starts strictly from the packet's true source seed, and request–boundary equivalence holds across all allowed finite action words, future admission ordinals, and sites.

Theorem E.1 demonstrates that even across large analytic pipelines, an uneliminated residual η cannot be silently dropped, but serves directly as syntactic fuel for the next verified frame.

Code and Data Availability

The OB2 compile-time firewall and the OB12–OB13 proofs, direct consumers and receipts are available at H0mework commit `ffa5eeb70130b17db3873407d0cd10a218e45896`. The second-edition claim map identifies the Lean declarations, complete types and original source revisions. The reproduction guide gives the source recovery, build and dependency audit commands. Run `python3 tools/one_bit_v2.py entry` to verify the accepted selection. CP483 and CP485 record the original independent certifications. The first-edition map locates the previously accepted proofs for OB1–OB11.

Run `make check-whole-ledger-full` to rebuild OB3, OB5 and OB8–OB10 and `make check-first-release-full` for OB6–OB7. The toolchain is pinned to Lean 4.33.0 and Mathlib v4.33.0.

Shared process theorems cite Process Core, second edition, DOI 10.5281/zenodo.23297175; unified-field and universal-formation theorems cite the Physics Flagship, third edition, DOI 10.5281/zenodo.24100339 (second edition DOI 10.5281/zenodo.23297179); and whole-ledger payment theorems cite Whole-Ledger Accounting, v1, DOI 10.5281/zenodo.23437454; the first edition of the present paper is archived at DOI 10.5281/zenodo.23437456. Original code is licensed under Apache-2.0; manuscript text and original vector figures are licensed under CC BY 4.0.

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